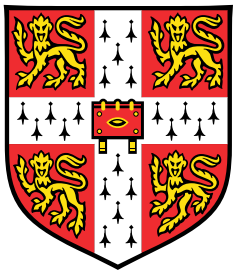
Physics-informed Neural Networks for predicting the road cracks



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Declaration

I hereby declare that except where specific reference is made to the work of others, the contents of this dissertation are original and have not been submitted in whole or in part for consideration for any other degree or qualification in this, or any other university. This dissertation is my own work and contains nothing which is the outcome of work done in collaboration with others, except as specified in the text and Acknowledgements. This dissertation contains fewer than 65,000 words including appendices, bibliography, footnotes, tables and equations and has fewer than 150 figures.

Xiaofang Wen
July 2026

Abstract

This is where you write your abstract ...

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Chapter 1

Introduction

1.1 Background and Motivation

Road pavements deteriorate under traffic loading, environmental exposure and material ageing. Cracking is one of the most common visible signs of this deterioration. Once cracks open at the surface, water can enter the pavement structure and weaken the lower layers. This can accelerate further damage and lead to potholes or more severe structural failure.

Road maintenance is also a major financial problem. The 2026 ALARM survey reported an estimated one-off carriageway repair backlog of £18.62 billion for local roads in England and Wales. The same survey estimated that 12 years would be needed to clear this backlog under current conditions [4]. It also reported that local authorities filled approximately 1.9 million potholes in the previous year, at an estimated cost of £149.3 million [4].

The importance of timing is also reflected in pavement preservation practice. FHWA describes pavement preservation as the application of the right treatment to the right pavement at the right time, with quality materials and construction [17]. Preservation treatments are intended to slow deterioration and extend service life before reconstruction becomes the only option. This shows why maintenance planning depends not only on the current pavement condition, but also on how deterioration is expected to progress.

This timing problem motivates two linked tasks: identifying the current crack state and predicting how it may grow. If maintenance is carried out too late, damage may have already spread through the pavement structure. If maintenance is carried out too early, limited resources may be spent on sections that are not yet at high risk. Inspection can help identify where cracking is present and what condition it represents, but maintenance planning also requires a prediction of how that state is likely to evolve. A better understanding of crack-growth behaviour can therefore support more targeted maintenance decisions, provided that

the prediction model is linked to the relevant material, loading and inspection information. Figure 1.1 illustrates this timing problem.

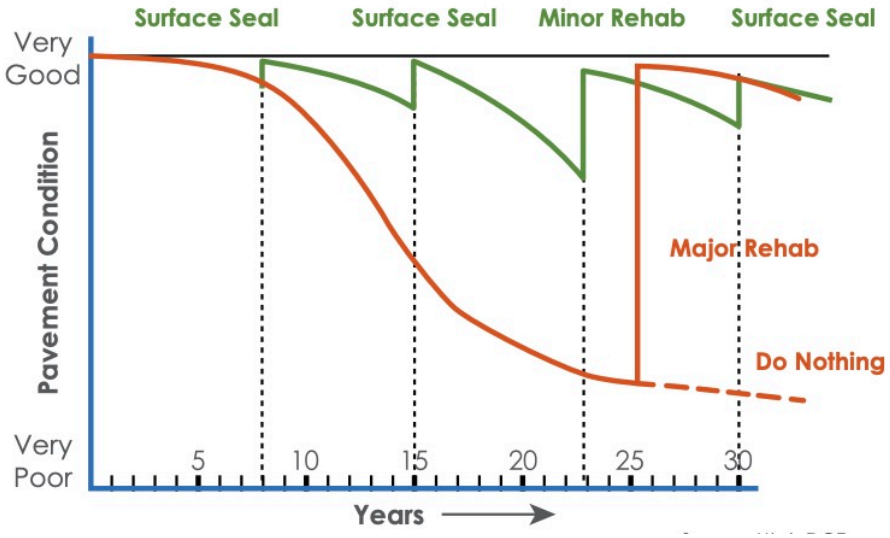


Fig. 1.1 Conceptual pavement deterioration and preservation timing. The figure illustrates the FHWA pavement preservation principle that the right treatment should be applied to the right pavement at the right time. Delayed intervention may allow pavement condition to decline until more costly rehabilitation is required. Adapted from FHWA and Utah DOT [17].

1.2 State of Practice

The timing problem described above is usually addressed in practice through condition monitoring and pavement management. Road authorities need repeated and comparable information about pavement condition before they can identify which sections are deteriorating, decide where maintenance is needed and prioritise limited budgets. In the UK, road condition data are used by local authorities to monitor, plan and prioritise highway maintenance [14]. Pavement management systems can also use condition histories, treatment rules and deterioration assumptions to support forward planning. The prediction target in these systems is usually the future performance or maintenance need of a road section or network. Examples include future condition scores, deterioration states, remaining service life and treatment timing. These predictions support asset management, but they do not directly describe the local crack path or crack-growth rate of an individual crack.

Condition assessment provides the observations used in this process. Traditional methods such as the Pavement Condition Index record visible distress by type, severity and extent [5]. Automated road-monitoring systems extend this process to larger road networks. In the UK,

SCANNER collects road-surface measurements and feeds them into pavement management systems such as UKPMS to support maintenance planning [13, 52]. More recent PAS 2161 guidance moves local road-condition monitoring towards approved survey technologies and a five-category condition format [14, 51]. These methods make road-condition data more consistent and easier to compare across locations.

Current practice can provide visible distress information and coarse prediction for maintenance planning. However, this prediction usually concerns the future condition or treatment need of a road section rather than the local evolution of an individual crack. It does not directly predict how a specific crack will extend, how fast it may grow, or which mechanical state variables control that growth. The gap addressed in this report is therefore the gap between condition-level pavement assessment and local fracture-state prediction.

1.3 Research Problem

The gap matters because crack growth is controlled by more than the visible crack recorded during inspection. A surface crack is only the observable part of a wider fracture process. Two cracks with similar surface length or width may evolve differently because of differences in loading history, material condition, layer stiffness and environmental exposure. These quantities are not fully observable from routine inspection. The prediction problem is therefore also an identification problem. In this report, these unobserved quantities are treated as hidden fracture-state quantities that need to be represented before local crack-growth prediction can become physically meaningful.

Physics-based fracture modelling provides one way to represent such hidden variables. In phase-field fracture modelling, cracks are described by a continuous damage variable rather than by an explicitly tracked sharp crack [39]. This allows crack initiation, propagation and branching to be represented without prescribing a crack path in advance. Many phase-field fracture models are implemented using finite-element methods (FEM) [7, 39, 1]. However, FEM does not remove the prediction gap by itself. It remains a forward numerical model that requires the geometry, material properties, boundary conditions and loading history to be specified, whereas these quantities are not normally available from routine pavement inspection. In this report, FEM is used as a controlled numerical reference for the selected phase-field crack-growth formulation.

Existing research addresses different sides of this problem, but not the complete bridge required here. Computer-vision and inspection methods can provide increasingly detailed descriptions of surface crack geometry, but they do not infer the mechanical fields that govern future propagation. Pavement deterioration models support network-level and section-level

asset planning, but their prediction targets are usually condition indices, deterioration states, remaining service life or treatment timing rather than the path and rate of an individual crack. Phase-field FEM can represent damage, strain-energy and history fields within a prescribed fracture problem, but it is not directly an inspection-based prediction method. Physics-informed deep learning (PIDL) offers a possible route for embedding fracture physics into a learning framework, yet most existing controlled fracture demonstrations remain closer to monotonic or quasi-static brittle failure than to cyclic fatigue degradation (close to repeated traffic loading). The research problem is therefore how to develop a prediction method that is physically meaningful but also suitable for later connection to inspection data.

As a first step toward this goal, we need to verify whether learning-based methods can handle the underlying physics under known numerical conditions. This report therefore studies physics-informed deep learning (PIDL) as a possible bridge, focusing on a controlled numerical benchmark. The research question is: to what extent can PIDL reproduce the fatigue-life response and hidden fracture-state evolution of a controlled FEM phase-field fatigue benchmark, and where does the formulation fail?

1.4 Research Design and Report Structure

The long-term motivation of this research is inspection-informed crack-growth prediction for pavement maintenance. However, the completed first-year work addresses a narrower prerequisite problem. It studies a controlled numerical fracture benchmark, not a real road-pavement validation case. The benchmark uses known geometry, material parameters, boundary conditions and loading history so that the behaviour of a physics-informed deep learning (PIDL) formulation can be compared with a finite-element reference.

The objectives of the report are:

1. to review current methods for pavement crack inspection and crack-growth prediction and physics-informed fracture modelling;
2. to identify the gap between visible crack inspection outputs and hidden fracture-state variables required for prediction;
3. to define a controlled phase-field fatigue benchmark with a finite-element reference solution;
4. to formulate a physics-informed learning framework for phase-field crack-growth prediction;

5. to evaluate whether the PIDL formulation can reproduce both global fatigue life and field-level quantities;

6. to identify the technical limitations of the current formulation and define the future work needed before moving towards pavement calibration.

The report is organised into five chapters. Chapter 1, *Introduction*, introduces the inspection-informed prediction motivation, the practice gap and the research design.

Chapter 2, *Literature Review*, reviews inspection outputs, pavement deterioration prediction and physics-informed fracture modelling. Inspection methods are reviewed only as the observed information available to a future prediction framework. The chapter then considers empirical, data-driven, physics-informed and fracture-based prediction methods, before summarising the gap between visible inspection outputs and hidden fracture-state evolution.

Chapter 3, *Controlled Phase-field FEM Reference and PIDL Formulation*, defines the controlled numerical benchmark, the phase-field fatigue reference model and the PIDL formulation used in the completed first-year work.

Chapter 4, *Diagnostic Development of the FEM–PIDL Fatigue Benchmark*, evaluates the PIDL formulation against the FEM reference. It reports both scalar fatigue-life behaviour and field-level diagnostics, and identifies the main formulation mismatch.

Chapter 5, *Research Proposal and Future Work*, uses these diagnostic findings to define the next research questions, work packages, deliverables and risks needed before the method can move towards pavement-material calibration and inspection-informed prediction.

Chapter 2

Literature Review

This chapter reviews the literature behind the research problem introduced in Chapter 1. It first reviews pavement crack inspection methods, because inspection defines what can be observed directly from the road surface or from structural survey data. It then reviews crack-growth prediction methods, from pavement-level deterioration models to physics-based and physics-informed fracture modelling.

2.1 Crack Inspection

Crack inspection is reviewed here because it defines the information available at the start of a crack-growth prediction problem. It identifies where cracks are visible, how severe they appear and how they are distributed across the road surface. These outputs are valuable for maintenance planning, but they are not the same as the hidden mechanical fields that govern future crack propagation. This section therefore reviews manual condition surveys, image-based segmentation, three-dimensional surface sensing and structural inspection in terms of the outputs they provide and their limitations for local crack-growth prediction. Figure 2.1 summarises the inspection categories and the types of outputs they can provide.

2.1.1 Manual and Automated Pavement Condition Surveys

Manual condition surveys provide a structured way to record visible pavement distress. Standards such as ASTM D6433 define procedures for identifying distress type, severity and extent, and for converting these observations into the Pavement Condition Index (PCI) [5]. This gives engineers a standardised procedure for describing pavement condition. However, manual surveys are limited by time and subjectivity. They are labour-intensive, and distress

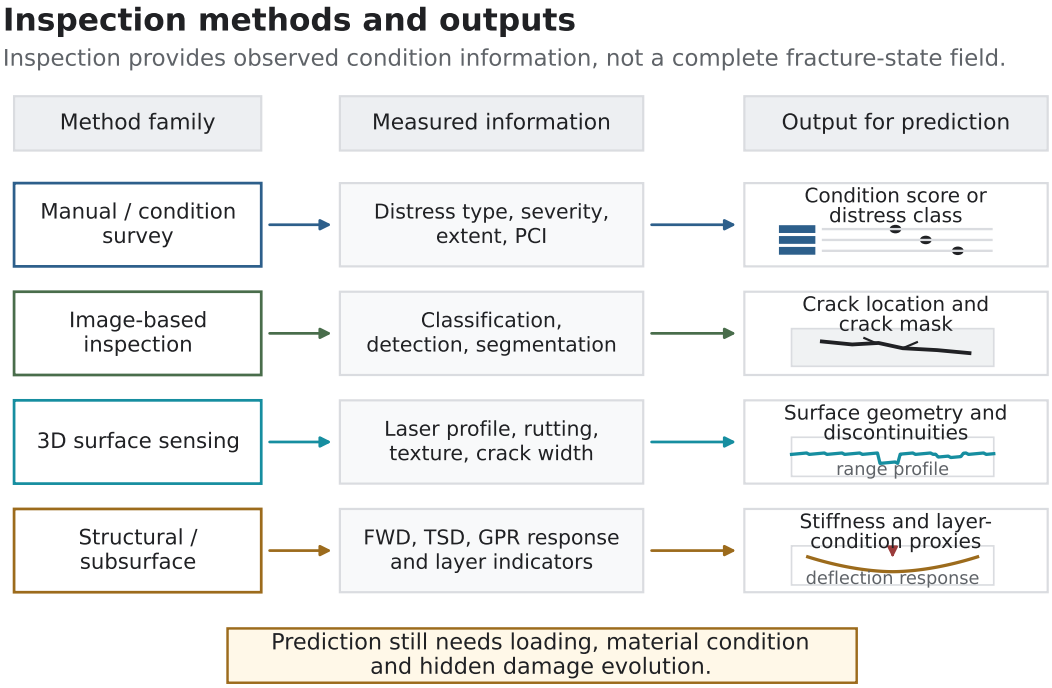


Fig. 2.1 Overview of pavement crack inspection methods and outputs.

type or severity may vary between inspectors [6, 35]. This makes frequent network-level inspection difficult if the process relies only on manual assessment.

Automated and semi-automated surveys address part of this problem. In the UK, road condition monitoring has been supported by systems such as SCANNER and UKPMS. SCANNER surveys use traffic-speed measurements of road surface condition, while UKPMS provides a framework for processing condition data and supporting maintenance planning [13, 52]. More recent guidance under PAS 2161 indicates a shift towards approved technologies for local road-condition monitoring. In this context, approved technologies are survey methods, sensing systems or data-processing tools that have been assessed as suitable for reporting road condition. These technologies may include image-based survey systems, three-dimensional surface measurement systems and other automated inspection methods. Their outputs are compared with engineer-assessed condition categories and reported using a five-category condition format. This is intended to support more consistent network-level reporting and maintenance planning [14, 51]. These survey systems support asset management decisions. They help authorities identify sections that require investigation or treatment. Their main output, however, is still a description or classification of current condition. This defines the observable information available for later prediction.

2.1.2 Image-based Crack Detection and Segmentation

Image-based inspection is widely used because visual data are relatively easy to collect. Vehicle-mounted cameras, mobile phones and unmanned aerial vehicles can all provide images for detecting pavement distress [3, 55]. Early methods used thresholding, edge detection, morphology and hand-crafted texture features [40, 54, 26, 25]. These methods show that visible crack pixels and crack geometry can be extracted from surface images, but they were sensitive to lighting, shadows, pavement texture and surface contamination [55, 57].

Deep learning has improved crack detection and segmentation. Object detection models can locate cracks or road damage using bounding boxes. Semantic segmentation models can classify each pixel as crack or background. Datasets such as Crack500 [55], DeepCrack [57] and RDD2022 [3] have supported this development by providing labelled images for training and benchmarking.

Different image-based models produce different types of outputs. A classification model indicates whether an image patch contains a crack or not. A detection model gives an approximate crack location with a bounding box. A segmentation model marks the crack pixels and can provide a binary or probabilistic crack mask to estimate the crack features. Figure 2.2 illustrates the difference between image-level classification, object detection and pixel-level segmentation outputs. However, image-based inspection remains a surface observation.

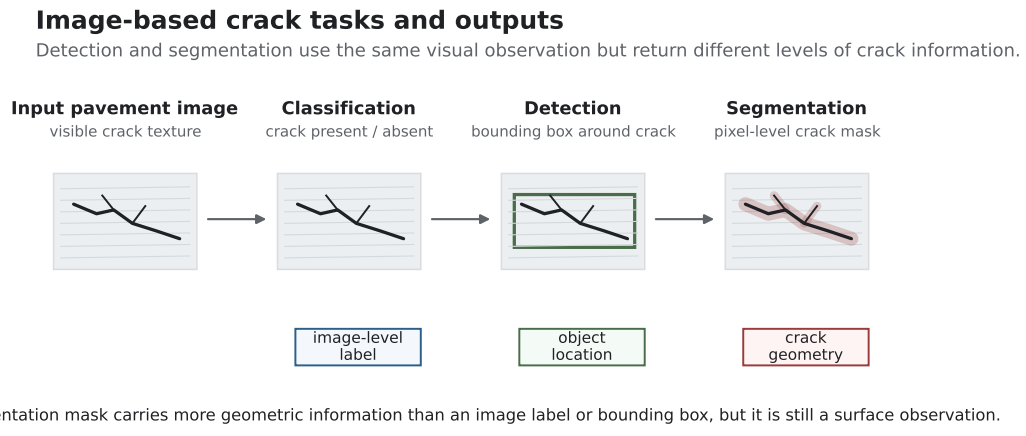


Fig. 2.2 Schematic comparison of image-based crack inspection tasks and outputs. Classification gives an image-level label, detection gives a bounding box or object location, and segmentation gives a crack mask with more geometric information.

2.1.3 Three-dimensional Surface Sensing and Structural Inspection

Three-dimensional surface sensing and laser profilometry also provide measurement of the road surface as a geometric profile rather than only as image appearance as shown in Figure 2.3. Systems such as the Laser Crack Measurement System (LCMS) combine intensity information with three-dimensional range data, so that surface distress can be described in terms of both appearance and shape [47]. This is useful because some crack-related features are geometric. A surface profile can provide information about crack width, surface discontinuity and local texture, which may be lost or simplified in a two-dimensional image or a binary crack mask [47]. These outputs provide richer observations for later prediction.

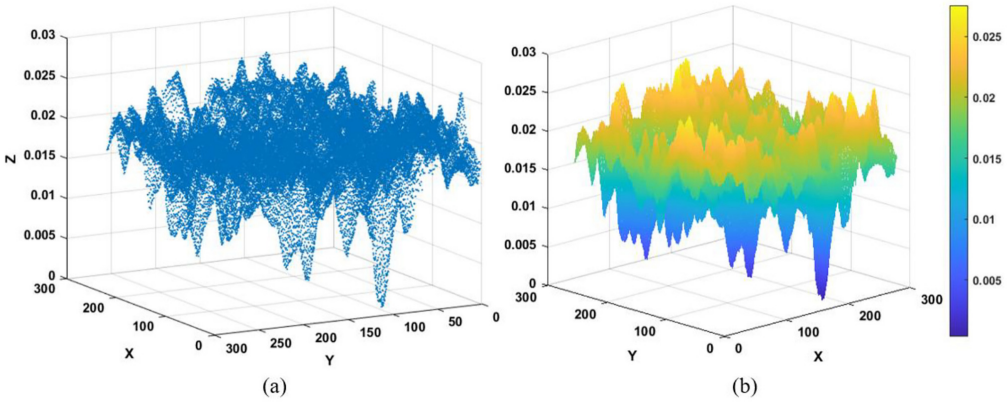


Fig. 2.3 Example three-dimensional pavement crack profile. Panel (a) shows a point-cloud profile and panel (b) shows the corresponding meshed surface profile. The example illustrates how a crack can appear not only as an image feature but also as a local geometric discontinuity or depression in the road-surface profile. Source: Pan et al. [41].

Structural and subsurface inspection methods measure pavement response or internal layer features rather than only surface appearance. Falling Weight Deflectometer (FWD) testing measures the deflection to an applied load and can be used to infer layer stiffness or structural capacity [18]. Traffic Speed Deflectometer (TSD) systems aim to collect deflection-related information at traffic speed [28]. Ground Penetrating Radar (GPR) can provide information about layer thickness, voids and other subsurface features [15]. These methods extend inspection beyond visible crack shape by adding indirect indicators of structural condition. However, they still give indirect indicators rather than a full local fracture-state field. A model is still needed to link these observations to hidden damage evolution.

2.1.4 Inspection Outputs and Their Limits

Inspection methods provide different levels of observed crack information. Manual and automated condition surveys usually provide distress classes, severity levels, condition scores or maintenance categories. Image-based detection and segmentation can provide crack locations, masks and geometric descriptors such as length, width, orientation, branching and connectivity, depending on image quality and post-processing. Three-dimensional surface sensing adds surface-profile information such as texture, discontinuity and local depression, while structural or subsurface inspection can provide indirect indicators of layer stiffness, structural capacity or internal anomalies. These outputs are valuable because they define the observed crack condition and provide practical inputs for maintenance assessment. Repeated surveys can also show visible changes in crack pattern over time.

However, these outputs are not the same as the local fracture-state variables required for crack-growth modelling. Crack propagation also depends on loading, material condition and the local damage process around the crack. These factors are not fully measured by routine inspection. For this reason, inspection can provide part of the initial observable state, but a predictive model is still needed to estimate how damage will develop after that point.

2.2 Crack Growth Prediction

Pavement deterioration and crack-growth prediction extend the inspection outputs reviewed above from observed condition to future change. Inspection records the crack or pavement condition at the time of observation. Prediction estimates how that condition may change in the future. In pavement maintenance, this can support decisions about treatment timing, risk and resource allocation [16]. Prediction methods do not all work at the same level of detail. Some methods predict pavement-level condition indicators, some predict distress quantities, and more mechanistic methods attempt to model the local damage process that drives crack initiation and propagation. This section uses the following four levels to organise the literature: empirical and statistical deterioration models, data-driven prediction, physics-informed prediction and fracture-based local crack-growth modelling.

2.2.1 Empirical and Statistical Deterioration Models

Empirical and statistical models are widely used in pavement management because they are simple and practical. They usually use existing pavement records to estimate future condition. These records may include pavement age, traffic loading, climate exposure,

Table 2.1 Observed inspection outputs and the hidden fracture-state information not directly provided by each method.

Inspection route	Typical output	Value for crack-growth prediction	Limitation for this report
Manual and condition surveys	Condition scores, distress types, severity levels and maintenance categories.	Provide interpretable asset-condition information and support network-level maintenance planning.	Usually coarse and discrete; do not provide local fracture-state variables such as stress, damage history or crack-driving force.
Image-based crack detection	Crack masks, crack pixels, crack length, width, density and surface geometry.	Provides visible crack geometry that can define the observed crack state.	Mostly surface-based; cannot directly infer subsurface damage, strain energy or future crack-driving conditions.
3D surface sensing	Surface texture, rutting, deformation, crack depth proxies and geometric profiles.	Adds richer geometric information than 2D images and can improve local surface characterisation.	Still mainly observes the pavement surface; hidden mechanical fields remain unobserved.
Structural or subsurface inspection	Deflection, layer stiffness proxies, layer condition indicators or subsurface anomalies.	Provides indirect evidence about structural capacity and internal deterioration.	Spatial resolution and interpretation can be limited; outputs are not direct phase-field fracture variables.
Combined inspection datasets	Integrated surface, structural and historical condition information.	Can provide stronger input features for future inspection-informed crack-growth models.	Still requires a model to connect observed features to hidden fracture-state evolution.

pavement structure and previous condition [16]. The output is often a condition index, a deterioration curve, a transition probability or a remaining-life estimate.

Many deterioration models do not track individual cracks. Instead, they predict how the overall condition of a road section changes over time or under accumulated traffic loading. Here, cracking is usually treated as one component of a broader pavement performance measure. This helps budget planning because they can predict how whole networks of roads will deteriorate when funding is limited.

Markov models represent deterioration as transitions between condition states. Condition ratings can be divided into discrete states, and transition probabilities can be used to estimate how likely a state is to move to another state in the future [8]. This treats deterioration as a probabilistic process rather than a single fixed curve.

Survival and hazard models take another statistical route. Instead of predicting a continuous condition score directly, they estimate the probability that a road section will either stay in good shape or drop below an acceptable standard over time. To make these predictions highly accurate, researchers use a massive US national database called Long-Term Pavement Performance (LTPP), which has tracked real-world highways for decades. By analysing this long-term data, survival modelling has been used to study pavement life and deterioration risk under different traffic, structural and environmental conditions [24].

Mechanistic-empirical methods include more engineering information. They first use a structural model to calculate pavement responses, such as stress or strain. Empirical transfer functions are then used to estimate distress or fatigue life [2]. This gives a closer link to pavement mechanics than purely statistical models, but the final prediction still depends on calibrated empirical relationships.

These models are useful for pavement management systems. They can be applied across road networks, require relatively simple inputs and support maintenance prioritisation. Their main limitation for this report is their prediction target. They usually predict section-level condition, deterioration risk or remaining life. They do not explain how a specific crack grows, and they do not provide local fracture variables such as material damage or strain-energy history.

2.2.2 Data-driven Prediction Models

Data-driven models use historical data to learn patterns between pavement inputs and future condition. These inputs may include pavement age, traffic history, climate variables, images and previous condition records, while the outputs are usually pavement-level performance indicators or distress quantities. In crack-related applications, such models have been used to predict distress categories, remaining useful life, future condition indices or wheel-path cracking [16, 50, 11, 36]. Their advantage is flexibility, because they can represent non-linear relationships that are difficult to encode in simple statistical equations.

However, purely data-driven models face a challenge in physical consistency. Because they only learn correlations from historical data, they can produce physically inconsistent predictions, especially when applied outside the distribution of their training data. This limitation demonstrates that data-driven predictions cannot rely on patterns alone, but they need to constrain predictions by physical principles.

2.2.3 Physics-informed Deterioration Prediction

To address the physical limitations of pure machine learning, physics-informed prediction integrates engineering domain knowledge directly into data-driven frameworks. By doing so, it reduces the black-box nature of machine learning and encourages predictions that are more consistent with engineering knowledge.

Recent studies show two main ways to do this. One route is rule-based constraints by adding engineering-informed constraints to a data-driven model. This approach enforces overall trend consistency without solving physical equations. For instance, Taheri et al. [50] constrained a deep learning model for wheel-path cracking by adding mathematical penalties to the loss function that guarantee monotonic behaviour. In this case, the rule forces the model to make sure the cracking strictly increases as road age and traffic volume grow.

The second route is equation-based constraints, which include a deterioration equation directly in the learning process. Li et al. [32] introduced a Physics-Informed Neural Network (PINN) that incorporates a fuzzy partial differential equation (PDE) governing road deterioration into the training loss. Here, the network uses automatic differentiation to ensure its predictions mathematically satisfy the underlying differential equations across all training scenarios. However, its predicted quantities are still pavement performance indicators, like PCI or RDI, rather than a local fracture-state field. Figure 2.4 shows the PINN structure used in this pavement-level fuzzy-PDE prediction setting.

For the present report, the key limitation is that these pavement-level physics-informed models still predict overall performance indicators or deterioration trends rather than local fracture-state fields. The review therefore turns next to fracture models and computational solvers, where damage, strain energy and history variables are represented explicitly.

2.2.4 Fracture Modelling and Computational Solvers

The previous subsections focused mainly on pavement-level prediction, such as the Pavement Condition Index (PCI), roughness or overall cracking quantities [16, 50]. These outputs are useful for network asset management, but they are too coarse to explain how an individual crack grows. Local crack growth depends on the mechanical state around the crack tip, including strain energy, fracture energy and material damage [42, 9, 39]. This subsection therefore reviews fracture models and computational solvers that can describe local crack-growth behaviour.

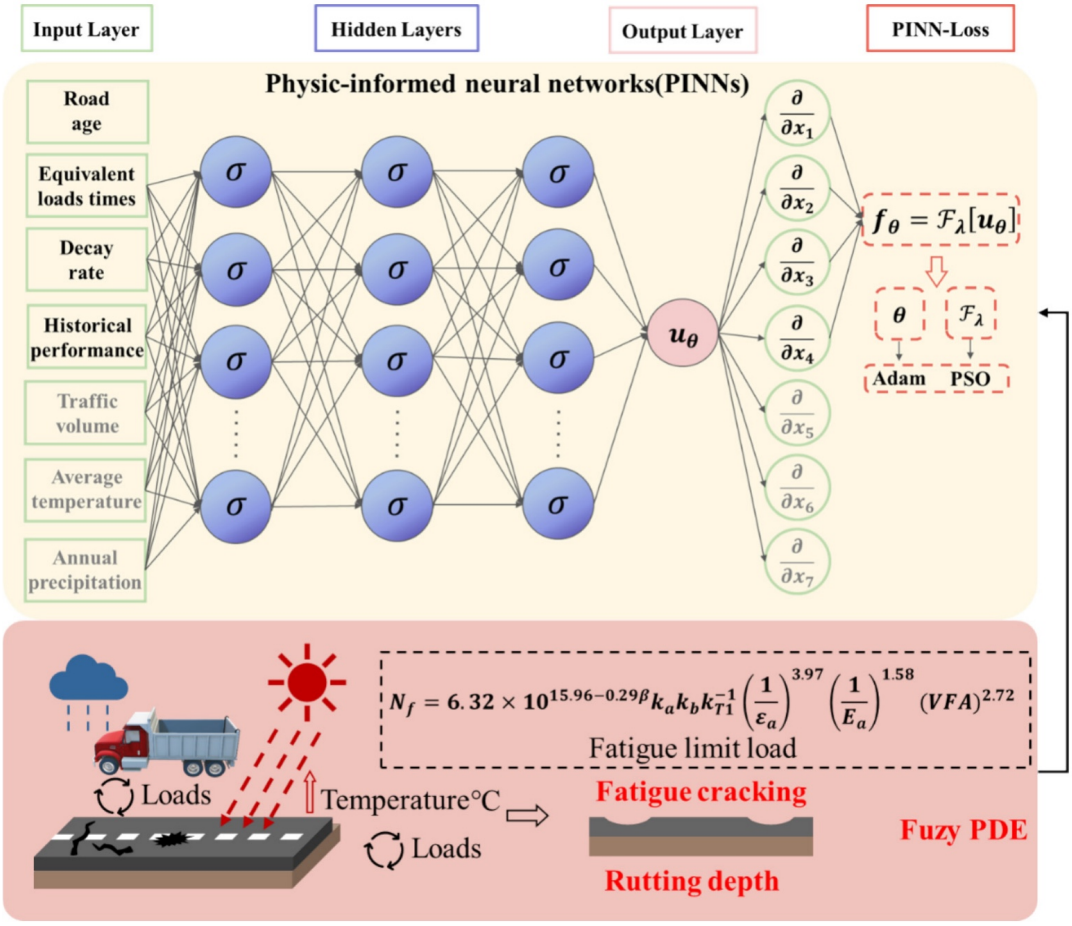


Fig. 2.4 PINN schematic for pavement performance prediction with a fuzzy PDE constraint. The model links pavement-level inputs, neural-network prediction, automatic differentiation and a physics-informed loss for PCI or RDI prediction. Source: Li et al. [32].

Fracture Models as Local Physical Descriptions

Classical fracture and damage models provide several ways to describe local crack-growth behaviour. Linear elastic fracture mechanics (LEFM) and Paris-type fatigue laws relate crack growth to quantities such as the stress intensity factor range [42, 30]. Cohesive zone models (CZM) describe the fracture process through traction–separation laws and fracture energy [49, 12]. Simplified viscoelastic continuum damage (VECD) models represent asphalt fatigue through internal damage variables and material response functions [53, 31].

These models demonstrate that crack propagation depends on hidden mechanical states rather than visible surface geometry alone. However, their practical requirements differ. LEFM and Paris-law models require explicit crack geometry and local loading information. CZM models require a calibrated traction–separation law and often depend on how the

crack path is represented in the mesh. VECD models require material-specific fatigue calibration from destructive tests. Collectively, these requirements are difficult to satisfy using routine non-destructive pavement inspection alone. This data and topological gap motivates modelling approaches that can represent damage as a continuous field, rather than only as an explicitly tracked crack path or a calibrated material degradation curve.

Phase-field fracture provides one such approach. Instead of tracking a sharp crack surface explicitly, it represents fracture using a continuous damage variable that changes smoothly between intact and broken material states [19, 7, 39, 1]. This allows the model to represent complex topological features, such as crack initiation, propagation, branching and coalescence, without explicitly remeshing the domain as the crack path changes [37]. Phase-field methods have also been applied to asphalt concrete and related pavement-material fracture problems [29]. For this report, the phase-field framework provides the underlying governing equations for crack growth. Table 2.2 compares these fracture-modelling routes by the type of damage state they represent and their relevance to this report.

Limits of Finite-element Phase-field Solvers

Many phase-field fracture models are solved using finite-element methods [29]. FEM can provide detailed displacement and damage fields, but it also has important limits. First, each new geometry, loading case or material setting normally requires a new numerical solve. This makes repeated forward simulation expensive when many scenarios need to be tested.

Second, FEM is most straightforward when used as a forward solver. It is effective when the material parameters, loading conditions and boundary conditions are known. It is less convenient when the aim is to infer hidden damage, material properties or loading history from incomplete inspection data. Such inverse problems usually require repeated calls to the forward solver inside an optimisation loop, which increases the total computational cost.

Third, real pavement applications involve uncertainty, partly because the available data are sparse or incomplete. Material properties, layer conditions, traffic loading and environmental exposure are not known exactly. Field inspections also provide only partial observations of the pavement state. A deterministic FEM simulation requires these quantities to be specified before the solve. Testing many possible parameter values can be computationally expensive, especially when the phase-field model itself is already costly to run.

Table 2.2 Fracture modelling routes and their representation of damage.

Model family	How damage or cracking is represented	Progress offered by phase-field fracture	Limitation for this report
Linear elastic fracture mechanics	Represents crack growth through sharp cracks and crack-tip quantities, such as stress intensity factors or energy release rates [42].	Phase-field fracture does not require explicit crack-tip tracking. Crack initiation, growth and branching can emerge from energy minimisation over a continuous damage field.	LEFM is difficult to apply when crack topology, branching or distributed damage evolves in a complex domain.
Cohesive zone models	Represents fracture through traction–separation laws along predefined or inserted cohesive interfaces.	Phase-field fracture is less dependent on predefined crack paths. Cracks can evolve through the domain as a continuous field on a fixed mesh.	CZM can be powerful when the crack path is known, but less flexible when arbitrary crack initiation or branching is expected.
Continuum damage / S-VECD models	Represents deterioration through averaged internal damage variables, often linked to stiffness degradation or fatigue damage.	Phase-field fracture provides a full-field damage variable, so the spatial position and topology of the crack can be represented explicitly.	Averaged damage models are useful for material characterisation, but they do not directly provide local crack geometry or crack-path evolution.
Finite-element phase-field fracture	Represents cracking through a continuous phase-field or damage variable coupled with displacement and fracture energy.	Provides the reference mechanics used in this report, including local damage, strain energy and history-dependent crack-driving variables.	Computationally expensive and requires fine spatial resolution near the diffuse crack zone.

Learning-based Solvers for Fracture Prediction

These limitations of FEM motivate learning-based solvers. Physics-informed neural networks (PINN) and related learning methods use governing equations, boundary conditions, residuals or energy terms to constrain the learning process [46, 27].

Phase-field fracture prediction is a challenging setting for these methods because crack propagation creates sharp, localised gradients near the crack tip and involves irreversible, history-dependent damage evolution. Strong-form residual PINNs can be difficult to train in such localised zones because they rely on pointwise PDE residuals and higher-order derivatives. For this reason, this report focuses on the variational and energy-minimisation branch, which smooths local gradients through total-energy integration, along with its solver extensions and operator-learning surrogates.

This energy-based route, exemplified by the deep Ritz method, aligns naturally with phase-field fracture. Because crack growth is governed by the balance between elastic energy release and fracture dissipation, the neural network can be trained to minimise a total energy functional rather than only pointwise PDE residuals. This functional couples displacement, damage and fracture history, making the approach suitable for phase-field fracture modelling [48]. Goswami et al. [21] proposed a transfer-learning-enhanced PINN for phase-field fracture modelling, where the network minimises the variational energy across consecutive loading steps and uses the previous load step to initialise the next one. Figure 2.5 shows the related deep Ritz structure used by Manav et al. [38].

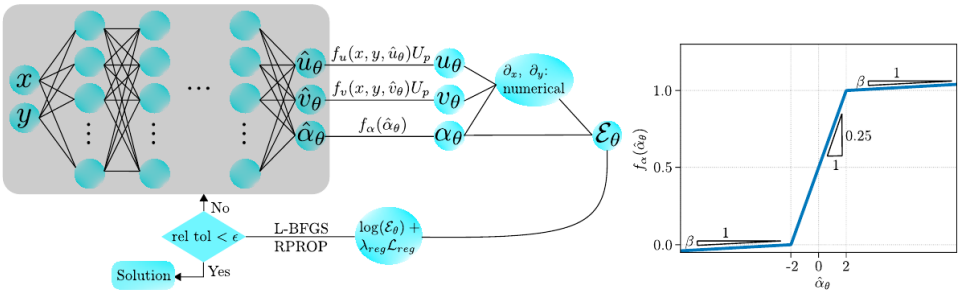


Fig. 2.5 Deep Ritz scheme for phase-field fracture. The network maps spatial coordinates to displacement and phase-field variables, and training minimises an energy-based loss. Source: Manav et al. [38].

Manav et al. [38] used a deep Ritz formulation to model phase-field fracture processes such as crack propagation, kinking, branching and coalescence. This study is the closest methodological baseline for the present report. Recent work has extended this variational learning route to more specialised phase-field settings, including anisotropic brittle crack propagation [44]. These studies show that energy-based neural solvers can reproduce controlled phase-field fracture benchmarks, but they mainly address monotonic or quasi-static brittle fracture. They do not directly resolve fatigue-driven crack growth under repeated loading, which is central to pavement deterioration.

Several extensions have been proposed to improve the robustness and efficiency of learning-based phase-field solvers. Domain-decomposed variants, such as extended variational PINNs, introduce local subdomains and interface constraints to improve approximation near the crack path [10]. Length-scale-aware PINN studies address another important issue: the phase-field length scale controls the width of the diffuse fracture zone and therefore affects the required spatial resolution. Lian et al. [33] introduced length-scale decoupling degradation functions to reduce this numerical burden, while Yu et al. [56] used PINNs

for inverse length-scale identification and forward phase-field fracture modelling. These studies show that phase-field learning methods are being extended beyond simple forward benchmarks into parameter identification and specialised solver design.

Comparative studies further show that phase-field PINN performance can be sensitive to activation functions, load steps and neural-network parameter choices [20]. Therefore, these solvers should be evaluated against reliable numerical reference solutions rather than assumed to be intrinsically robust. They are still not direct solutions to fatigue-driven pavement crack-growth prediction. Figure 2.6 gives an example of a recent deep Ritz workflow for anisotropic phase-field fracture, which illustrates how these solver extensions are organised.

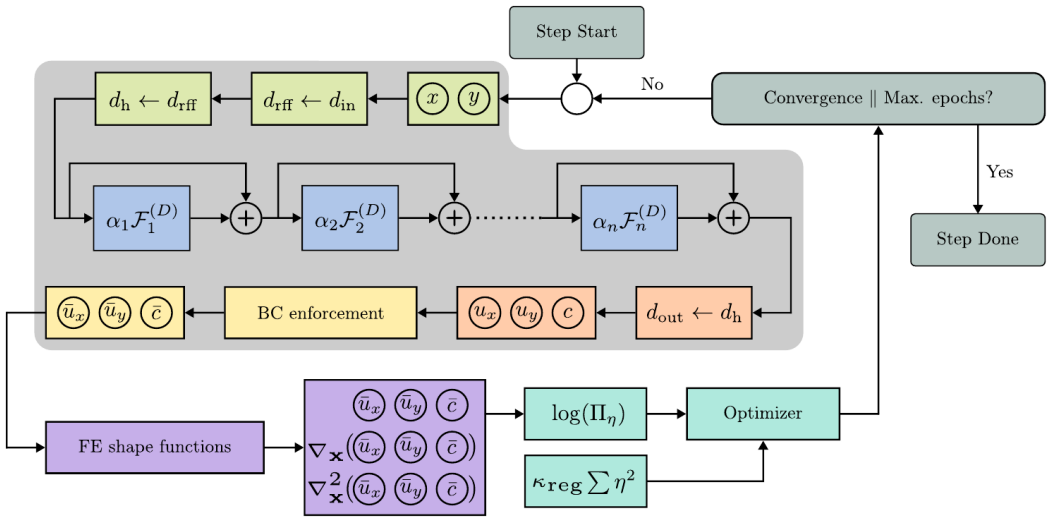


Fig. 2.6 Deep Ritz workflow for anisotropic phase-field fracture. The workflow shows coordinate input, neural-network field prediction, boundary-condition enforcement, energy evaluation and optimisation. Source: Plungė et al. [44].

Operator-learning methods and supervised surrogates form a related but distinct branch. Instead of solving one fracture problem at a time, they aim to learn mappings from input functions or parameters to output fields. Goswami et al. [22] introduced a physics-informed variational DeepONet for predicting crack paths from initial defect configurations. Domain-agnostic neural operators have also been developed for irregular geometries and topological changes [34]. However, these methods should be positioned carefully. Some operator-learning fracture demonstrations are based on other fracture frameworks, such as peridynamics, rather than phase-field fracture. Recent benchmark studies also show that robust surrogate modelling for phase-field fracture remains an open problem, because crack topology, high-frequency damage features and metric choice can strongly affect evaluation

reliability [23]. They are therefore treated as future routes for fast repeated prediction rather than as direct evidence for the current PIDL solver.

The reviewed studies show that physics-informed learning for phase-field fracture has moved beyond simple forward benchmarks toward more complex scenarios. However, most existing energy-based PIDL studies still focus on monotonic or quasi-static brittle fracture, rather than fatigue-driven crack growth under repeated loading which is closer to real pavement deterioration. Table 2.3 summarises the learning routes reviewed here and clarifies their role in this report.

Table 2.3 Learning-based fracture solvers and their role in this report.

Learning route	Representative sources	Input and output	Loss or constraint	Role in this report
Strong-form residual PINN	Raissi et al. [46]; Karniadakis et al. [27].	Coordinates and boundary or data points to solution fields.	Pointwise PDE residuals with boundary and data losses.	General physics-informed learning background; not the main route used in the completed work.
Variational / energy-based learning, including Deep Ritz	Samaniego et al. [48]; Goswami et al. [21]; Manav et al. [38]; Plungė et al. [44].	Coordinates, boundary or loading conditions to displacement and phase-field variables.	Total potential energy or variational fracture functional.	Methodological basis for PIDL; existing studies mainly use controlled brittle or quasi-static settings.
Domain-decomposed variational learning	Chakraborty et al. [10].	Subdomain coordinates to local displacement and damage fields.	Energy losses with interface constraints between subdomains.	Targets local approximation near crack paths; adds solver complexity.
Length-scale-aware or inverse phase-field PINN	Lian et al. [33]; Yu et al. [56].	Coordinates, observations or length-scale settings to damage fields or identified length scale.	Phase-field loss with length-scale treatment or inverse-identification terms.	Relevant to phase-field sensitivity and inverse problems; still controlled phase-field examples.
Operator learning and surrogates	Goswami et al. [22]; Liu et al. [34]; Hamdi and Lejeune [23].	Initial defects, geometry, loading or simulation parameters to crack paths or fields.	Operator-learning or supervised surrogate loss, sometimes with physics-informed terms.	Future route for fast repeated prediction; not direct evidence for the current PIDL formulation.

2.3 Summary: Research Gap and Position of This Report

This chapter reviewed the literature on pavement inspection, pavement-level deterioration prediction and physics-informed fracture modelling. Inspection methods can provide observed crack and pavement-condition information, including distress type, severity, surface crack geometry and, in some cases, structural or subsurface indicators. Pavement deterioration models can then use such information to support section-level or network-level maintenance

planning by predicting condition indices, deterioration states, distress quantities or treatment timing. Although valuable for broad asset management, these established methodologies operate at a coarse operational scale and cannot directly capture local fracture-state variables that govern physical crack propagation. Local crack growth depends on hidden mechanical quantities such as phase-field damage, tensile strain-energy density, history variables, fatigue degradation and active crack-driving fields. A modelling framework is therefore needed to connect observed crack-condition information with hidden fracture-state evolution.

Phase-field fracture mechanics provides a full-field representation of crack evolution within a prescribed fracture model, and finite-element implementations provide a suitable numerical reference for such problems. However, FEM simulations remain forward numerical solvers: they require geometry, material properties, boundary conditions and loading histories to be specified, and they can be computationally expensive for repeated crack-growth simulations. Physics-informed deep learning (PIDL) offers a possible alternative, particularly through variational and energy-minimisation formulations. However, existing neural fracture solvers are limited to monotonic or quasi-static brittle configurations rather than cyclic fatigue degradation. This leaves a methodological gap between what routine inspection observes, what pavement-level deterioration models predict, and what local fatigue-fracture modelling requires.

The position of this report is therefore deliberately limited. It does not attempt to predict real pavement cracks directly from inspection data. Instead, it first tests whether a PIDL formulation can reproduce a controlled FEM phase-field fatigue benchmark at both scalar and field levels. This controlled comparison is a prerequisite diagnostic step: if the learning formulation cannot reproduce the hidden damage, history and crack-driving fields in a known numerical setting, it cannot yet be used as evidence for pavement-level crack-growth prediction.

Chapter 3

Completed Work

This chapter defines the controlled numerical benchmark used to test the PIDL formulation. The benchmark is not a pavement validation case. It is a two-dimensional phase-field fatigue problem with prescribed geometry, material parameters, boundary conditions and cyclic loading. The purpose is to create an aligned setting in which the FEM reference and PIDL approximation can be compared at both scalar and field levels.

The chapter first defines the SENT benchmark geometry, loading and normalised parameters. It then states the phase-field fatigue formulation used by both solvers, including the elastic degradation, damage energy, fatigue-history update and fatigue degradation law. The finite-element reference and PIDL implementation are then described separately, followed by the alignment checks used before the Chapter 4 diagnostic comparison.

3.1 Purpose and Scope of the Controlled Benchmark

The completed work uses a controlled two-dimensional phase-field fatigue benchmark. The benchmark is a single-edge notched tension (SENT) specimen with prescribed geometry, a horizontal pre-crack, displacement-controlled cyclic loading and normalised material parameters. Its purpose is to test whether a PIDL solver can reproduce the response of a FEM reference under aligned numerical conditions.

The benchmark removes features of real pavement cracking. The material is treated as homogeneous, isotropic and linear elastic. The loading considered here is idealised cyclic tension rather than traffic loading, environmental cycling or mixed-mode pavement loading. The material parameters are normalised benchmark quantities rather than calibrated pavement properties. This normalisation is used to create a numerically controlled problem in which the displacement, damage, energy and history terms remain on manageable scales for PIDL optimisation. These assumptions make the problem suitable for diagnosing the

phase-field fatigue formulation, but they also mean that the results should not be interpreted as pavement-level crack prediction.

3.2 Benchmark Problem Definition

The controlled benchmark is a two-dimensional single-edge notched tension (SENT) problem. Figure 3.1 defines the geometry, pre-crack, boundary conditions and loading used throughout the chapter. The computational domain is a unit square region $\Omega = [-0.5, 0.5]^2$. A horizontal pre-crack is introduced along $y = 0$ from $x = -0.5$ to $x = 0$, giving a normalised initial crack length $a_0/L = 0.5$. The phase-field length scale is set to $l_0 = 0.01$ as a fixed benchmark regularisation length and its modelling role is discussed in Section 3.3.3.

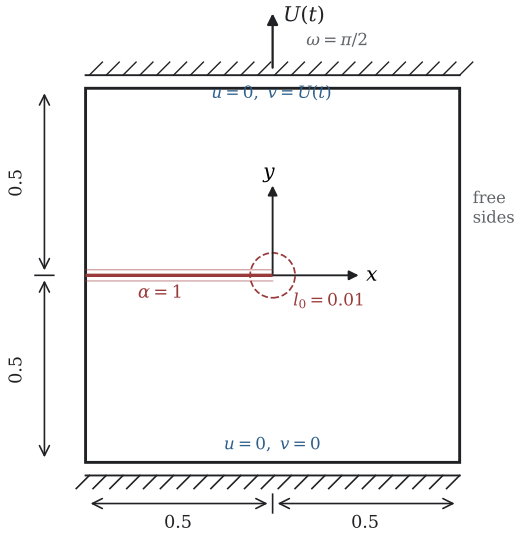


Fig. 3.1 Schematic of the controlled SENT benchmark. The figure defines the domain, coordinate system, horizontal pre-crack, boundary conditions, cyclic displacement loading, phase-field length scale and normalised geometric quantities used in the FEM–PIDL comparison.

The displacement boundary conditions are prescribed on the horizontal edges. The bottom edge ($y = -0.5$) is fixed in both displacement components, so $u = 0$ and $v = 0$. On the top edge ($y = 0.5$), the horizontal displacement is fixed and the vertical displacement is prescribed, so $u = 0$ and $v = U(t)$. The left and right side edges are traction-free. The loading is displacement-controlled cyclic tension with maximum displacement ($U_{\max}$), minimum

displacement ($U_{\min} = RU_{\max}$), and loading ratio

$$R = \frac{U_{\min}}{U_{\max}}.$$

The fatigue benchmark uses $R = 0$, so the prescribed displacement cycles between zero and $U_{\max}$.

The benchmark is written in dimensionless numerical units by taking the specimen width L and Young's modulus E as reference scales, with $L = 1$ and $E = 1$. The main dimensionless groups are the normalised crack length a_0/L, the phase-field length ratio l_0/L, the Poisson's ratio ν, the cyclic loading ratio $R = U_{\min}/U_{\max}$, the normalised damage-energy coefficient w_1, and the fatigue-history threshold α_T. These values should therefore be interpreted as benchmark parameters, not as calibrated pavement material properties. Table 3.1 summarises the adopted values.

The coefficients w_1 and α_T require particular care because they control damage and fatigue degradation in the normalised model. In a dimensional phase-field formulation, the damage-energy density is governed by the fracture toughness G_c and the phase-field length l_0. In the normalised implementation used here, this role is represented by the benchmark coefficient w_1, so w_1 is not treated as a measured pavement fracture toughness. Similarly, α_T is the threshold on the normalised fatigue-history scale at which fatigue degradation begins. The modelling roles of w_1, l_0 and α_T are discussed below in Section 3.3.3.

Table 3.1 Normalised benchmark parameters used in the controlled SENT fatigue problem.

Quantity	**Value**	**Interpretation**
L	1.0	Reference specimen width; the domain is $[-0.5, 0.5]^2$.
E	1.0	Normalised Young's modulus; not a calibrated pavement modulus.
ν	0.3	Poisson's ratio used in the linear elastic model.
a_0/L	0.5	Normalised initial pre-crack length.
l_0/L	0.01	Normalised phase-field length scale controlling the diffuse crack width.
$R = U_{\min}/U_{\max}$	0	Displacement loading ratio; the cycle unloads to zero displacement.
w_1	1.0	Normalised damage-energy coefficient corresponding to the G_c/l_0-type scale in the adopted formulation.

α_T	0.5	Normalised fatigue-history threshold for the onset of fatigue degradation.

3.3 Phase-field Fatigue Model

3.3.1 Energy Function and Model Variables

This section introduces the phase-field model. The purpose is to define the core variables, energy terms and history quantities shared by both the FEM reference and the PIDL implementation.

The main idea of the phase-field model is to represent a crack using a continuous damage variable instead of a sharp crack surface. This damage variable is denoted by α, where $\alpha = 0$ represents intact material and $\alpha = 1$ represents fully damaged material. Crack growth is therefore described as the evolution of a continuous damage field across the domain.

The energy functional is useful for showing the overall structure of the model. The phase-field energy used in this benchmark follows the standard variational phase-field fracture structure [19, 7, 39, 9]:

$$\begin{aligned}\mathscr{E}^n(\mathbf{u},\alpha;\bar{\alpha}^n) = &\underbrace{\int_\Omega \left[g(\alpha)\psi^+(\varepsilon(\mathbf{u})) + \psi^-(\varepsilon(\mathbf{u}))\right] d\Omega}_{E_{\text{el}}} \\ &+ \underbrace{\int_\Omega f_{\text{fatigue}}(\bar{\alpha}^n)\frac{w_1}{c_w}\left[w(\alpha) + l_0^2|\nabla\alpha|^2\right] d\Omega}_{E_{\text{d}}}.\end{aligned} \tag{3.1}$$

Equation 3.1 structures the rest of this section. The first underbraced term, E_{el}, couples displacement, strain energy and damage. The second underbraced term, E_{d}, penalises the creation and spatial variation of phase-field damage, with fatigue degradation applied through $f_{\text{fatigue}}(\bar{\alpha})$. In this damage-energy term, $w(\alpha)$ penalises local damage creation, while $l_0^2|\nabla\alpha|^2$ penalises abrupt spatial changes in the damage field. Together, these terms produce the diffuse damage band in the phase-field model, with its width controlled by the phase-field length scale l_0. The phase-field length scale l_0 controls the width of the diffuse damage zone and affects the apparent strength of the model. In a material-calibrated pavement application, it would need to be chosen using fracture tests, mesh-resolution checks and information about the physical fracture process zone. Here, $l_0/L = 0.01$ is fixed as a benchmark choice so that FEM and PIDL use the same regularisation length. ALl the symbol definitions can be found in Table 3.2.

Table 3.2 Symbols used in the phase-field fatigue energy functional.

Symbol	**Definition**
Ω	Two-dimensional SENT computational domain.
$\mathbf{u} = (u, v)$	Displacement field.
α	Phase-field damage variable, with $\alpha = 0$ intact and $\alpha = 1$ fully damaged.
$\bar{\alpha}$	Accumulated fatigue-history variable controlling fatigue degradation.
$\varepsilon(\mathbf{u})$	Infinitesimal strain tensor computed from the displacement field.
ψ^+	Positive crack-driving elastic-energy density.
ψ^-	Compressive or non-driving elastic-energy density.
$g(\alpha)$	Elastic degradation function.
$w(\alpha)$	Local damage-density function; $w(\alpha) = \alpha$ for AT1.
c_w	AT1 damage-density normalisation constant.
l_0	Phase-field length scale.
w_1	Normalised damage-energy coefficient.
$f_{\text{fatigue}}(\bar{\alpha})$	Fatigue degradation function multiplying the damage-energy term.

Crack growth can be understood as an energy trade-off. If the material stays intact, the tensile part of E_{el} remains stored. If α increases, the model pays a damage cost through E_{d} and the irreversibility treatment, but it can release part of the tensile elastic energy through $g(\alpha)\psi^+$. Fatigue makes damage cheaper by reducing the effective damage-energy cost through $f_{\text{fatigue}}(\bar{\alpha})$. The minimisation therefore favours damage when this route gives a lower total energy. This is a qualitative interpretation of the energy minimisation problem, not a separate crack-growth criterion. The main quantities in this competition are summarised in Table 3.3.

Table 3.3 Main quantities and qualitative crack-growth roles in the phase-field fatigue energy used in Chapter 3.

Symbol	Name	Effect on damage growth	Feedback when α increases
u	Displacement field	Sets the strain and therefore the elastic-energy densities. Larger tensile strain raises the crack-driving part of the energy.	Damage changes the degraded stiffness locally, so the displacement and strain fields can redistribute around the damaged zone.

α	Phase-field damage variable	Represents the diffuse crack state, from intact material at $\alpha = 0$ to fully damaged material at $\alpha = 1$.	A larger α lowers $g(\alpha)$, raises the local damage-density contribution and is prevented from healing by the history treatment.
E_{el}	Elastic-energy term	Stores the mechanical energy that can be reduced by damage in tensile regions.	Increasing α reduces the degraded tensile part of this term through $g(\alpha)\psi^{+}$.
E_{d}	Damage-energy term	Sets the energetic cost of local damage and spatial damage gradients.	Increasing α usually raises the local and gradient contributions, unless fatigue degradation has reduced the effective cost.
$g(\alpha)$	Elastic degradation function	Controls how strongly tensile elastic energy is degraded in damaged material.	Since $g(\alpha) = (1-\alpha)^2$, larger damage lowers the tensile stiffness contribution.
ψ^{+}, ψ_{raw}	Positive crack-driving elastic-energy density	Contains the tensile and deviatoric elastic energy that can drive crack growth.	It is not directly degraded by α, but the strain field can change as damage redistributes the response.
ψ^{-}	Negative volumetric elastic-energy density	Keeps compression in the elastic energy without letting it directly promote crack growth.	It is controlled without multiplication by $g(\alpha)$.
$\bar{\alpha}$	Accumulated fatigue-history variable	Stores positive fatigue-driving activity and controls fatigue degradation.	It is carried forward with loading history; it is not reduced when the current damage or load decreases.
$f_{\mathrm{fatigue}}(\bar{\alpha})$	Fatigue degradation function	Reduces the effective damage-energy cost after the fatigue-history threshold is exceeded.	It depends on $\bar{\alpha}$ rather than directly on α, but damage can affect future history increments through the active driver.
w_1, c_w	Damage-energy scale and normalisation	Scale and normalise the cost of creating phase-field damage.	They are fixed benchmark parameters in this chapter.
l_0	Phase-field length scale	Controls the width of the diffuse damage zone and weights the gradient contribution.	It stays fixed, but it controls how sharply an increase in α can localise.

The energy functional also explains how the displacement field and the damage field are coupled. The displacement field changes the strain energy, while the damage field degrades part of that strain energy and contributes to the damage-energy term. For a given loading state n and accumulated fatigue history $\bar{\alpha}$, the reference phase-field problem is therefore

written as an energy minimisation problem, subject to the prescribed boundary conditions and the history treatment:

$$(\mathbf{u}^n, \alpha^n) = \arg\min_{\mathbf{u},\alpha} \mathscr{E}^n(\mathbf{u}, \alpha; \bar{\alpha}^n). \tag{3.2}$$

3.3.2 Elastic-energy Term E_{el}: Degradation and Split

The elastic degradation function of this benchmark is

$$g(\alpha) = (1-\alpha)^2. \tag{3.3}$$

This function reduces the active elastic-energy contribution as damage grows. This is a modelling choice commonly used in phase-field fracture formulations, rather than a material law calibrated for pavement cracking. Both FEM and PIDL solvers use the same degradation assumption to ensure the test consistency. In numerical practice, a residual-stiffness parameter k is sometimes added to $g(\alpha)$ to prevent complete loss of stiffness as α approaches one.

The infinitesimal strain components are

$$\varepsilon_{11} = u_{,x}, \qquad \varepsilon_{22} = v_{,y}, \qquad \varepsilon_{12} = \frac{1}{2}(u_{,y} + v_{,x}). \tag{3.4}$$

Here a comma denotes partial differentiation, for example $u_{,x} = \partial u/\partial x$.

The elastic energy density is separated into three contributions: a positive volumetric part, a negative volumetric part and a deviatoric part. The positive volumetric and deviatoric contributions form the crack-driving energy density and are multiplied by the degradation function $g(\alpha)$. The negative volumetric contribution represents compression and is controlled without degradation, so compression is not allowed to directly promote crack growth. The positive and negative volumetric energy densities are written as

$$\psi_V^+ = \frac{1}{2}K\langle 3\varepsilon_m\rangle_+^2, \qquad \psi_V^- = \frac{1}{2}K\langle -3\varepsilon_m\rangle_+^2, \tag{3.5}$$

where

$$K = \lambda + \frac{2}{3}\mu, \qquad \varepsilon_m = \frac{\varepsilon_{11}+\varepsilon_{22}}{3}. \tag{3.6}$$

Here λ and μ are the Lamé parameters corresponding to the chosen Young's modulus and Poisson's ratio.

The deviatoric contribution is included in the degraded positive part of the energy:

$$\psi_{\mathrm{dev}} = \mu\left[(\varepsilon_{11}-\varepsilon_m)^2 + (\varepsilon_{22}-\varepsilon_m)^2 + (-\varepsilon_m)^2 + 2\varepsilon_{12}^2\right]. \tag{3.7}$$

The raw crack-driving energy density is therefore defined as the sum of the positive volumetric contribution and the deviatoric contribution:

$$\psi^{+} = \psi_{\text{raw}} = \psi_V^{+} + \psi_{\text{dev}}. \tag{3.8}$$

The degraded active driver is

$$\psi_{\text{active}} = g(\alpha)\psi_{\text{raw}}, \tag{3.9}$$

The active driver will be used for the fatigue-history update explained in 3.3.4.

With these definitions, the code evaluates the energy terms elementwise using element areas A_e. In this implementation, the elastic energy is

$$E_{\text{el}} = \sum_e A_e \left[g(\alpha_e)\psi_{\text{raw},e} + \psi_{V,e}^{-} \right]. \tag{3.10}$$

Here e indexes the finite elements, A_e is the area of element e, and α_e denotes the element-mean phase-field value. The quantities $\psi_{\text{raw},e}$ and $\psi_{V,e}^{-}$ are the corresponding elementwise raw tensile and negative volumetric energy densities.

3.3.3 Damage-energy Term E_{d}: Damage Density and Fatigue Degradation

The damage-energy term contains the local damage density $w(\alpha)$, the phase-field gradient contribution $l_0^2|\nabla\alpha|^2$ and the fatigue degradation $f_{\text{fatigue}}(\bar{\alpha})$. The length scale l_0 sets the width of the diffuse damage band that represents a crack in the phase-field description. It is therefore a modelling and numerical regularisation parameter: a smaller l_0 gives a sharper damage transition, while a larger l_0 spreads the transition over a wider zone. The benchmark uses an AT1 regularisation for the crack-surface energy. In phase-field formulation, the crack is represented by a continuous damage variable α. The value $\alpha = 0$ corresponds to intact material and $\alpha = 1$ to fully damaged material. In the AT1 model, the local damage-density function is linear in α, with $w(\alpha) = \alpha$ for $0 \leq \alpha \leq 1$. In the AT1 model, the local damage-density function is linear within $0 \leq \alpha \leq 1$.

$$w(\alpha) = \alpha, \qquad c_w = \frac{8}{3}. \tag{3.11}$$

Here, c_w is the normalisation constant associated with the chosen crack-density function. It ensures that the regularised crack-surface energy is scaled consistently with the fracture energy. Together with a gradient penalty on $\nabla\alpha$, these terms allow the model to approximate a crack as a diffuse damage band whose width is controlled by the phase-field length scale

l_0. The AT1 form is a modelling choice within the controlled benchmark, not a pavement-calibrated material law [43]. Other crack-density functions are possible. For example, the AT2 model uses a quadratic form, $w(\alpha) = \alpha^2$ with different value of c_w. The present benchmark uses AT1 because it is simpler and gives a clearer damage response.

In the elementwise implementation, the phase-field damage energy is

$$E_{\mathrm{d}} = \sum_e A_e f_{\mathrm{fatigue}}(\bar{\alpha}_e) \frac{w_1}{c_w} \left[w(\alpha_e) + l_0^2 \left((\partial_x \alpha)_e^2 + (\partial_y \alpha)_e^2 \right) \right]. \tag{3.12}$$

Following the variational phase-field fatigue framework of Carrara et al. [9], fatigue is introduced by multiplying the damage-energy term by a degradation function of an accumulated fatigue-history variable. The benchmark uses

$$f_{\mathrm{fatigue}}(\bar{\alpha}) = \begin{cases} 1, & \bar{\alpha} \leq \alpha_T, \\ \left(\dfrac{2\alpha_T}{\bar{\alpha} + \alpha_T} \right)^2, & \bar{\alpha} > \alpha_T, \end{cases} \tag{3.13}$$

where $\bar{\alpha}$ is the accumulated fatigue-history variable and and α_T is the threshold below which no fatigue degradation is applied. For $\bar{\alpha} > \alpha_T$, the function decreases monotonically and reduces the effective damage dissipation as fatigue history accumulates as shown in Figure 3.2. The value $\alpha_T = 0.5$ is adopted as a normalised benchmark parameter in the controlled implementation. It is not calibrated from pavement fatigue tests. Its role is to define the onset of fatigue degradation within the controlled numerical comparison. Sensitivity of the results to α_T is therefore a necessary part of future material calibration.

The degradation factor enters the phase-field formulation through the damage-energy term, rather than by directly scaling the stress. It therefore reduces the local fracture resistance as fatigue history accumulates. In dimensional notation, this can be interpreted as an effective fracture-energy scale,

$$G_c^{\mathrm{eff}} = f_{\mathrm{fatigue}}(\bar{\alpha}) G_c. \tag{3.14}$$

In the normalised benchmark used here, the same effect appears through the factor $f_{\mathrm{fatigue}}(\bar{\alpha})$ multiplying the damage-energy coefficient in E_{d}.

3.3.4 Fatigue History Update and Fatigue Degradation

The fatigue history variable $\bar{\alpha}$ is different from the damage variable α. The damage variable α describes the current phase-field damage. The history variable $\bar{\alpha}$ records the accumulated fatigue-driving activity at each element. It is then used in $f_{\mathrm{fatigue}}(\bar{\alpha})$ to reduce the effective damage-energy term.

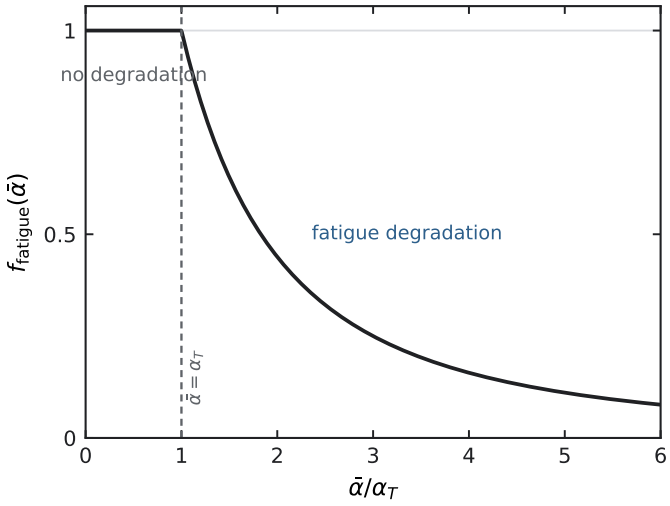


Fig. 3.2 Fatigue degradation law used in the controlled benchmark. The horizontal axis is the normalised fatigue-history variable $\bar{\alpha}/\alpha_T$, and the vertical axis is the degradation factor f_{fatigue} multiplying the damage-energy term. The threshold $\alpha_T = 0.5$ is an adopted normalised benchmark parameter, not a calibrated pavement fatigue threshold.

In a phase-field fracture model, fatigue history is driven by the change of active tensile-energy quantity ψ_{active} defined in Equation 3.9, not by the raw tensile energy ψ_{raw} alone. The update therefore accumulates only positive increases in this active driver:

$$\Delta\bar{\alpha} = \left\langle \psi^n_{\text{active}} - \psi^n_{\text{prev}} \right\rangle_+, \qquad \bar{\alpha}^{n+1} = \bar{\alpha}^n + \Delta\bar{\alpha}. \tag{3.15}$$

Here ψ^n_{prev} is the stored driver value from the last loading cycle used as the reference for the current increment. In this notation, ψ^n_{prev} is only the reference driver used to compute the increment; the accumulated fatigue history $\bar{\alpha}$ is carried forward through Equation 3.15. The Macaulay bracket keeps only positive changes, so unloading does not remove fatigue history that has already been accumulated.

3.3.5 Damage Irreversibility Penalty

In a phase-field fracture model, damage should not decrease once it has formed. Otherwise, the numerical solution would allow crack healing. This would be non-physical for the crack-growth problem considered in this report.

This issue is important because the fields are obtained by minimising an energy-based objective. Without an irreversibility treatment, the optimiser may reduce the objective by lowering damage that has already appeared. The implementation therefore adds an

irreversibility penalty as part of the optimisation objective to discourage the current damage field α from falling below the stored maximum α_{hist}.

The irreversibility penalty is written as

$$E_{\text{hist}} = \frac{1}{2} w_1 \gamma_{\text{ir}} \sum_e A_e \left\langle -(\alpha_e - \alpha_{\text{hist},e}) \right\rangle_+^2, \tag{3.16}$$

where $\alpha_{\text{hist},e}$ stores the previous maximum damage value in element e.

The factor γ_{ir} controls the strength of the irreversibility penalty. For the AT1 setting used in the implementation, it is defined as

$$\gamma_{\text{ir}}^{\text{AT1}} = \frac{27}{64\,\texttt{tol_ir}^2}. \tag{3.17}$$

This factor is not fitted from the FEM data. It is a numerical penalty coefficient used by the AT1 formulation. The inverse-square dependence on `tol_ir` reflects the quadratic form of the healing penalty: reducing the tolerated healing scale increases the penalty strength.

3.4 FEM Reference Implementation

This section describes the generation of the FEM reference solution used to assess the PIDL slver. The reference is generated using the GRIPHFiTH implementation [37], which olves the adopted variational phase-field fracture problem using standard FEM degrees of freedom. The primary unknowns are the displacement field $\mathbf{u} = (u, v)$ and the phase-field damage variable α. For the fatigue extension, additional path-dependent quantities are are stored and updated, including the damage-history field α, the accumulated fatigue-history variable $\bar{\alpha}$, the raw tensile driver ψ_{raw} and the degraded active driver ψ_{active}.

Because fatigue analysis is path-dependent, we only use 'committed' states for our comparison. This ensures we are comparing against finalized, stable results rather than the temporary trial data generated during the solution process. At the start of a step, the FEM solver reads the previously committed damage field and history variables from the previous step. After the displacement increment is applied, the solver uses a staggered procedure: the displacement problem is solved using the current damage field, and the phase-field problem is then solved using the current tensile energy and the previously committed history variables. During phase-field assembly, fatigue-related quantities are evaluated at Gauss points. These quantities become part of the accepted FEM state only after the Newton and staggered iterations satisfy their convergence criteria. After convergence, the damage history is committed by updating the stored maximum damage field, and the fatigue-history variables

are refreshed for the next step. Algorithm below summarises the workflow for one FEM loading step.

FEM workflow for one loading step. The FEM update for one loading step can be summarised as follows:

1. **Read the committed old state.** Retrieve

$$S_{k-1} = \left\{ p_{\text{old}}, \bar{\alpha}_{\text{old}}, f_{\text{old}}, \psi_{\text{prev,old}} \right\},$$

   where p_{old} is the committed damage history, $\bar{\alpha}_{\text{old}}$ is the accumulated fatigue history, f_{old} is the fatigue degradation factor and $\psi_{\text{prev,old}}$ is the previous active driver.

2. **Apply the displacement increment.** Prescribe the displacement level for loading step k.

3. **Solve the coupled problem by staggered iteration.** Repeat the following operations until the displacement, phase-field and stagger residuals satisfy the convergence criteria:

   (a) solve the displacement problem using the current damage field;

   (b) compute the strain field and the undegraded tensile driver Y at Gauss points;

   (c) solve the phase-field damage problem using p_{old}, the old history variables and the current tensile driver;

   (d) at each Gauss point, compute the active driver

$$\psi_{\text{active},q} = g(d_q) Y_q;$$

   (e) compute the positive fatigue-history increment

$$\Delta\bar{\alpha}_q = \left\langle \psi_{\text{active},q} - \psi_{\text{prev,old},q} \right\rangle_+;$$

   (f) update the candidate accumulated fatigue history

$$\bar{\alpha}_{\text{new},q} = \bar{\alpha}_{\text{old},q} + \Delta\bar{\alpha}_q;$$

   (g) evaluate the candidate fatigue degradation

$$f_{\text{new},q} = f_{\text{fatigue}}\left(\bar{\alpha}_{\text{new},q}\right).$$

4. **Commit the converged state.** After the staggered solution has converged, update the committed damage history and fatigue-history variables:

$$p_{\text{old}} \leftarrow \max(p_{\text{old}}, p), \qquad \bar{\alpha}_{\text{old}} \leftarrow \bar{\alpha}_{\text{new}},$$

$$f_{\text{old}} \leftarrow f_{\text{new}}, \qquad \psi_{\text{prev,old}} \leftarrow \psi_{\text{active}}.$$

5. **Save the reference state.** Save the converged post-commit state S_k for diagnostic comparison in Chapter 4.

3.5 PIDL Formulation

This section describes how the physics-informed deep learning (PIDL) model approximates the controlled phase-field fatigue problem defined above. The method follows the mesh-supported Deep Ritz Method used by Manav et al. [37]: a neural network represents the unknown solution fields, and its parameters are trained by minimising an energy-based objective. The main extension in this report is the fatigue treatment, entering through the fatigue-history update and the fatigue-degraded damage-energy term.

3.5.1 Mesh-supported Deep Ritz Representation

In the Deep Ritz approach, the unknown solution fields are represented by a feed-forward neural network. In this report, these fields are the displacement field $\mathbf{u} = (u, v)$ and the phase-field damage variable α. The neural network therefore acts as a continuous approximation of the solution fields over the spatial domain.

For a given loading step, the transformed neural fields enter the phase-field energy:

$$\mathscr{E}_\theta^n = \mathscr{E}^n(\mathbf{u}_\theta, \alpha_\theta; \bar{\alpha}^n). \tag{3.18}$$

Here θ denotes the trainable network parameters. This equation links the phase-field problem to the PIDL solver.

This is different from a supervised field regressor. In the controlled baseline, the PIDL model does not learn from FE displacement or damage labels during training.

Although the solution fields are represented by a neural network, the controlled PIDL implementation is not fully mesh-free. Following Manav et al. [37], the domain is discretised with triangular elements for numerical integration and gradient evaluation. The neural network provides displacement and phase-field values at spatial coordinates, while elementwise

shape functions are used to compute field gradients and elementwise quadrature is used to approximate the energy integrals.

The mesh is therefore used to evaluate the PIDL loss, not to introduce finite-element degrees of freedom as the primary unknowns. The trainable unknowns remain the neural-network parameters. This mesh-supported Deep Ritz implementation keeps the neural-field representation, but uses a finite-element-style discretisation to evaluate the variational loss.

3.5.2 Energy-based Loss Function

The PIDL objective follows a regularised Deep Ritz form. The variational part is built from the elastic energy, the fatigue-scaled damage energy and the irreversibility penalty:

$$\mathscr{L}_{\text{variational}} = \log_{10}\left(E_{\text{el}} + E_{\text{d}} + \lambda_{\text{hist}} E_{\text{hist}}\right) + \lambda_w \sum_i W_i^2. \tag{3.19}$$

The first part of the loss is the regularised Deep Ritz objective. It contains the elastic energy, the fatigue-scaled damage energy and the irreversibility penalty. The second part is a neural-network regularisation term. Here $\mathscr{W}$ denotes the set of trainable network weights. The weight-regularisation term penalises large network weights during optimisation, but it is not part of the physical phase-field energy.

The base-10 logarithm is a numerical scaling step, following the Deep Ritz scaling used by Manav et al. [37]. It keeps the loss magnitude manageable when the energy changes by orders of magnitude during crack growth. Since the logarithm is monotonic, it preserves the ordering of the positive energy objective inside the brackets.

3.5.3 PIDL Architecture and Boundary Treatment

PIDL takes the spatial coordinates (x,y) as inputs and returns three raw outputs:

$$(x,y) \mapsto \left(\hat{u}_{\text{raw}}, \hat{v}_{\text{raw}}, \hat{\alpha}_{\text{raw}}\right). \tag{3.20}$$

The three raw outputs are transformed in two different ways. The displacement outputs are not used as free displacement fields. Instead, they are inserted into a boundary-condition ansatz so that the main Dirichlet conditions are satisfied by construction.

For the vertical loading branch, this trial space can be written as

$$\begin{aligned} u(x,y) &= \lambda \left[B_y(y)\hat{u}_{\text{raw}}(x,y) + L_y(y)\cos\theta\right], \\ v(x,y) &= \lambda \left[B_y(y)\hat{v}_{\text{raw}}(x,y) + L_y(y)\sin\theta\right], \end{aligned} \tag{3.21}$$

where $B_y(y) = (y - y_0)(y_L - y)$ is zero on the lower and upper boundaries, $L_y(y) = (y - y_0)/(y_L - y_0)$ is the imposed linear lift, λ is the loading amplitude and $\theta = \pi/2$ for the vertical SENT loading. In this case, Equation 3.21 enforces $v = 0$ on $y = y_0$, $v = \lambda$ on $y = y_L$ and removes the displacement correction on both horizontal boundaries. The displacement boundary treatment is therefore a hard trial-space constraint rather than a penalty term added to the loss.

By contrast, the phase-field damage output is transformed by a range function rather than by a boundary-condition ansatz:

$$\alpha = S_{\mathrm{ns}}(\hat{\alpha}_{\mathrm{raw}}). \tag{3.22}$$

The controlled implementation uses the nonsmooth sigmoid

$$S_{\mathrm{ns}}(z) = \begin{cases} \varepsilon_{\mathrm{tail}}(z+s), & z < -s, \\ \dfrac{z}{2s} + \dfrac{1}{2}, & -s \le z \le s, \\ 1 + \varepsilon_{\mathrm{tail}}(z-s), & z > s, \end{cases} \qquad s = 2, \quad \varepsilon_{\mathrm{tail}} = 10^{-3}. \tag{3.23}$$

The phase-field output is not passed through a standard sigmoid. A standard sigmoid would keep α strictly between zero and one, but its slope becomes very small near the bounds. This can make crack nucleation difficult during training because the optimiser receives little gradient information when α is close to zero or one.

Following Manav et al. [37], the raw network output is instead mapped using the piecewise transformation in Equation 3.23. The central branch maps the main range of the raw output into $0 \le \alpha \le 1$. The two outer branches keep small non-zero slopes, so the optimiser can still update the damage field near the bounds. This design keeps the damage field trainable near $\alpha = 0$ and $\alpha = 1$, but it does not impose an exact bound. Small values slightly below zero or above one are therefore treated as numerical artefacts of the output transformation, not as physically meaningful damage states.

3.5.4 Training and History Update

During training, the PIDL loop updates the damage history α_{hist} and fatigue history $\bar{\alpha}$ at each loading step. The damage history α_{hist} records the maximum damage reached so far. This update helps prevent the stored damage field from decreasing, while the optimisation still uses the soft irreversibility penalty described in Section 3.3.5. The fatigue history $\bar{\alpha}$ is updated from the degraded active tensile energy ψ_{active}, rather than the raw tensile energy ψ_{raw}.

A model may show reasonable agreement in raw tensile energy but still accumulate the wrong fatigue history if its degraded active driver is incorrect. For this reason, the training loop is evaluated not only by scalar crack timing, but also by the coupled evolution of the fields.

Figure 3.3 summarises this loading-step update sequence.

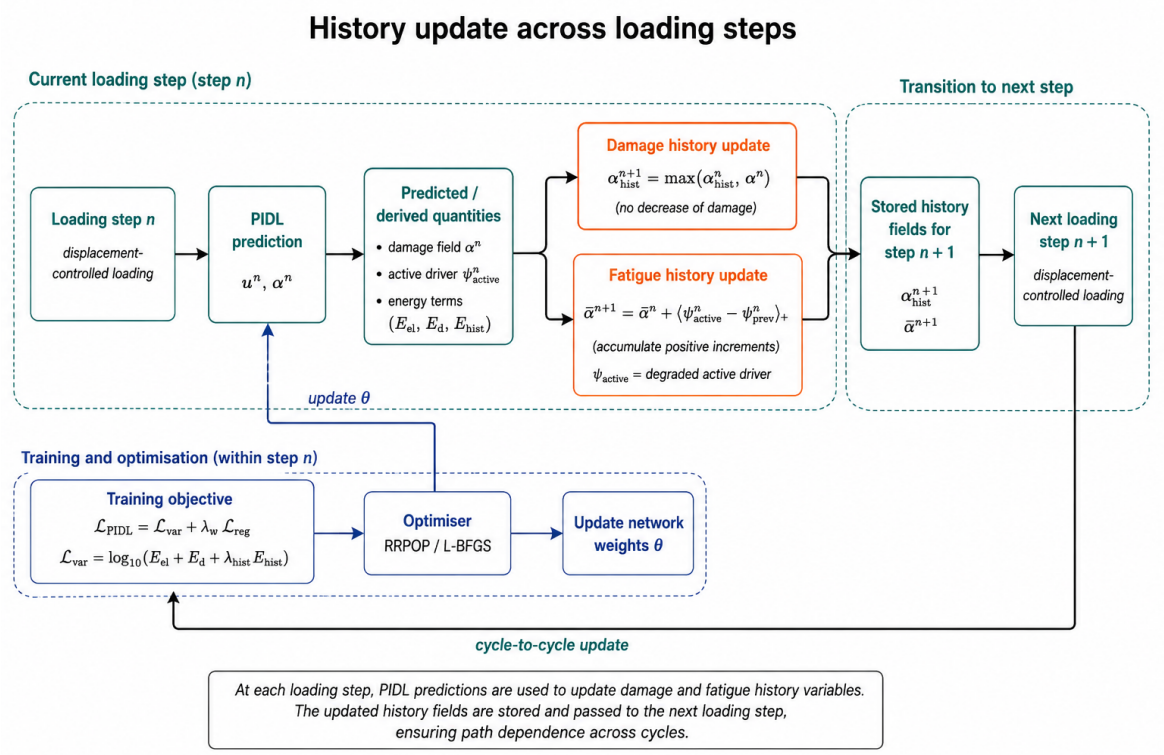


Fig. 3.3 History update sequence across PIDL loading steps. During loading step n, the network parameters are optimised using the stored damage history, accumulated fatigue history, fatigue degradation factor and previous driver. After optimisation and final field evaluation, the current damage field and active driver are used to refresh α_{hist}, $\bar{\alpha}$, f_{fatigue} and the next-step baseline. These refreshed fields are then passed to loading step $n+1$.

3.6 FEM–PIDL Alignment and Reproducibility Protocol

3.6.1 Model-level Alignment

The FEM reference and PIDL formulation are aligned at the level of the benchmark definition and the adopted energy terms, but they differ in the numerical representation of the unknown fields in the way the energy minimisation is carried out. The FEM reference represents **u**

and α through finite-element degrees of freedom and evaluates the discretised energy on the finite-element mesh. The PIDL formulation represents the same fields through a neural network and minimises an energy-like objective.

Table 3.4 Model-level alignment between the FEM reference and PIDL formulation.

Item	FEM reference	PIDL formulation
Benchmark definition	Controlled SENT geometry, horizontal pre-crack, displacement boundary conditions and cyclic loading history.	Same benchmark definition.
Material and model parameters	Uses the normalised parameters, phase-field length scale, elastic degradation, AT1 damage-energy term and fatigue degradation law defined above.	Same nominal parameter set and energy terms.
Primary fields	Displacement $\mathbf{u} = (u, v)$ and damage α represented by finite-element degrees of freedom.	Displacement $\mathbf{u}_\theta$ and damage α_θ represented by neural-network outputs.
Numerical evaluation	Finite-element mesh defines both approximation space and integration.	Mesh is used for integration and gradient evaluation, while trainable unknowns are neural-network parameters.
Role in comparison	Controlled numerical reference.	Learning-based approximation tested against the FEM reference.

3.6.2 Monotonic Sanity Check

Before the fatigue comparison, a monotonic alignment check is used to test the basic FEM–PIDL setup. This check uses the same SENT geometry and a comparable phase-field formulation, but it does not include cyclic fatigue-history accumulation.

This check has a limited role. It is not used to explain the fatigue results directly. Instead, it asks whether FEM and PIDL behave similarly in a simpler monotonic problem. Since this

monotonic problem follows the Deep Ritz benchmark of Manav et al. [37], the check helps verify the basic setup before the fatigue extension.

Both the FEM and PIDL solvers use the SENT specimen introduced in Section 3.2. The monotonic check uses displacement-controlled loading up to $U_{\max} = 0.2$. Figure 3.4 documents the meshes used in the numerical setup.

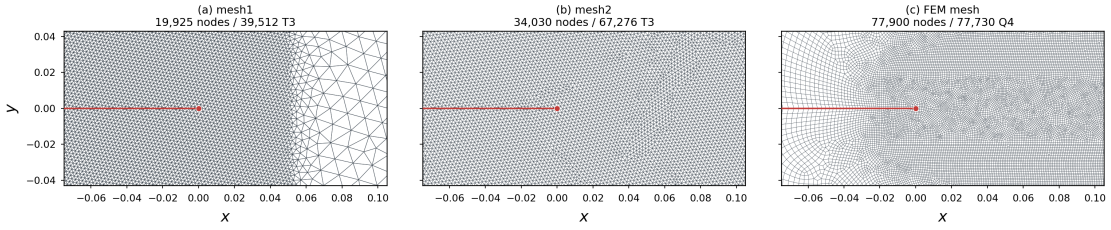


Fig. 3.4 Near-notch mesh comparison used to document the controlled numerical setup. The panels show `mesh1` with 19,925 nodes and 39,512 triangular elements for pretraining, `mesh2` with 34,030 nodes and 67,276 triangular elements for PIDL full training, and the original Abaqus FEM mesh with 77,900 nodes and 77,730 four-node quadrilateral elements. The red line marks the initial notch and the red point marks the notch tip.

The comparison focuses on the energy paths. To check the baseline alignment, these monotonic energy curves are overlaid directly onto the original Deep Ritz benchmark plot from Manav et al. [37], as shown in Figure 3.5.

The monotonic check is not the same problem as the selected fatigue benchmark because it uses monotonic rather than cyclic loading. It therefore checks only the basic phase-field setup alignment and does not test fatigue-history accumulation or fatigue degradation.

3.6.3 Cyclic Fatigue Alignment Check

The final fatigue comparison uses cyclic displacement-controlled loading with load ratio $R = 0$ and peak amplitude $U_{\max}$ from 0.08 to 0.12. These peak amplitudes are below the monotonic fracture transition window, which occurs around $U = 0.140$–0.145 in the alignment check. The fatigue case therefore tests crack growth under repeated sub-critical loading rather than immediate monotonic failure.

The diagnostic comparison uses both scalar and field-level quantities. Scalar quantities include fatigue life and crack-event timing. Field-level quantities include crack-tip position, phase-field damage α, fatigue history $\bar{\alpha}$, raw tensile driver ψ_{raw}, active degraded driver ψ_{active}, elastic energy E_{el}, damage energy E_{d} and incremental damage energy ΔE_{d}.

The reasons for adopting this aligned setup, and the effects of these choices on the FEM–PIDL comparison, are examined in Chapter 4.

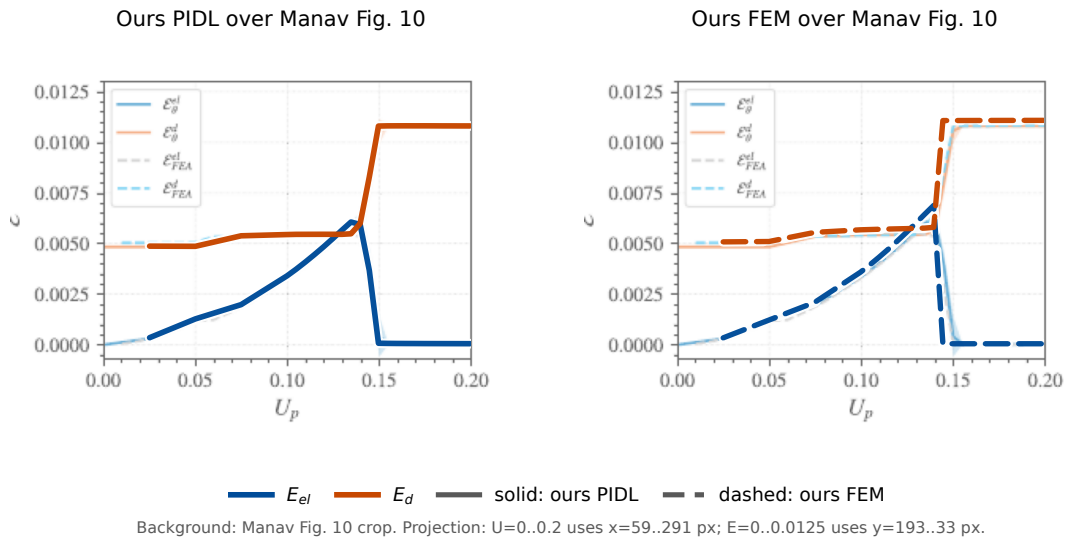


Fig. 3.5 Coordinate-aligned diagnostic overlay against the monotonic energy curves reported by Manav et al. [37]. The faded background is a crop of their Figure 10. The overlaid curves show the present PIDL eight-seed mean and the present GRIPHFiTH FEM monotonic reference projected onto the same axes. This overlay is used only to diagnose the monotonic baseline alignment; it is not fatigue validation or exact reproduction evidence for the finite-element reference in Manav et al. [37].

Initial State and History Variables

Because the fatigue benchmark is path-dependent, the initial state must be specified before the FEM–PIDL comparison can be interpreted. The initial crack is represented as a diffuse phase-field pre-crack rather than as a sharp geometric discontinuity. In the retained benchmark, the pre-crack is placed along $y = 0$ from $x = -0.5$ to $x = 0$, with the diffuse damage profile controlled by the phase-field length scale l_0. The same pre-crack definition is used for the FEM reference and the PIDL formulation.

At the start of the fatigue calculation, the accumulated fatigue-history field is set to zero. The stored previous driver field is also initialised consistently with the first loading state used in the cyclic update. This is necessary because the fatigue degradation function depends on the accumulated history $\bar{\alpha}$, and any mismatch in the initial history fields would change the subsequent fatigue-life response.

State-variable Storage and Update Audit

The fatigue benchmark is path-dependent, so the FEM–PIDL comparison depends not only on the final damage field but also on how the hidden state variables are stored, reduced and updated. The implementation audit in Table 3.5 is based on the current code semantics

rather than only on the continuum notation. This distinction is important because the FEM reference stores several fatigue variables at Gauss points, whereas the PIDL implementation stores the corresponding diagnostic quantities mainly at nodal or element level.

The audit also separates damage history from fatigue-energy history. In the current penalty-based fatigue path, the FEM damage-history quantity corresponding to the PIDL `hist_alpha` field is `p_field_old`, not `history_vars(:,:,1)`. The accumulated fatigue-history variable should instead be compared through `history_vars(:,:,2)` in FEM and `hist_fat` or `alpha_bar` in PIDL, after a stated GP-to-element reduction of the FEM field.

Table 3.5 State-variable storage, reduction and update audit for the FEM–PIDL fatigue benchmark. Node denotes nodal quantities, Elem denotes elementwise quantities, and GP denotes Gauss-point or quadrature-point quantities.

Physical quantity	**Role**	**FEM variable and location**	**PIDL variable and location**	**Recommended comparison level**	**Alignment status / audit note**
Displacement field, $\mathbf{u} = (u, v)$	Primary mechanical field used to compute strain and tensile energy.	`displ` or `u`; Node finite-element degrees of freedom.	Neural-network outputs `u`, `v`; evaluated at nodes.	Node, with the same boundary-condition convention.	Partly aligned. FEM uses finite-element degrees of freedom, whereas PIDL uses a neural-network trial space.
Current damage field, d or α	Phase-field damage variable representing the diffuse crack.	`p_field`; Node, interpolated to GP during assembly.	`alpha`; Node. Diagnostics also use `alpha_elem`.	Node for damage maps; Elem for field-ratio diagnostics.	Partly aligned after the reduction rule is stated.
Damage history	Enforces damage irreversibility through the committed maximum damage state.	`p_field_old`; Node, interpolated to GP for penalty checks.	`hist_alpha`; Node.	Node for max-history; GP/probe level for penalty diagnostics.	Max-history semantics are aligned after a committed step, but the penalty is evaluated at different spatial locations.
Irreversibility penalty	Penalises damage healing.	Checks $d_{\mathrm{GP}} - d_{\mathrm{old,GP}}$ at GP level.	Computes nodal α − `hist_alpha`, reduces to an element mean, then applies ReLU.	GP/probe level if strict alignment is required.	Not fully aligned. In general, $\mathrm{ReLU}(\mathrm{mean}(\Delta d)) \neq \mathrm{mean}(\mathrm{ReLU}(\Delta d))$.
Undegraded tensile driver, Y or ψ_{raw}	Raw tensile crack-driving energy before fatigue-history or degradation coupling.	`strain_en_undgr(element_GP)`; GP.	`psi_raw_elem`; Elem.	FEM GP-to-Elem reduction versus PIDL Elem.	Approximately aligned only after a stated reduction rule.
Active crack-driving field, ψ_{active}	Degraded driver used in the fatigue-history update.	$g(d_{\mathrm{GP}})Y_{\mathrm{GP}}$; GP.	`psi_active_elem` or `psi_plus_elem`; Elem or probe-reduced Elem.	FEM GP-to-Elem reduction versus PIDL Elem/probe reduction.	Critical mismatch location. FEM uses GP-level $g(d_q)Y_q$; PIDL uses reduced element diagnostics.

Accumulated fatigue history, $\bar{\alpha}$	Stores positive fatigue-driving increments and controls fatigue degradation.	`history_vars(:,:,2)`; GP.	`hist_fat` or `alpha_bar`; Elem.	FEM GP-to-Elem reduction versus PIDL Elem.	Correct fatigue-history pair to compare after FEM reduction is stated.
Previous fatigue baseline	Stores the previous active driver used to compute positive fatigue-history increments.	`history_vars(:,:,3)`; GP. Stores the degraded driver for the next update.	`psi_plus_prev`; Elem.	FEM GP-to-Elem reduction versus PIDL Elem, with state label.	Timing-sensitive. Saved PIDL `psi_plus_prev_elem` may already be the next-step baseline.
Fatigue degradation factor, f_{fatigue}	Multiplies the damage-energy term and reduces effective damage resistance.	`history_vars(:,:,4)`; GP.	`f_fatigue`; Elem.	FEM GP-to-Elem reduction versus PIDL Elem.	Approximately aligned after reduction, but FEM candidate f can affect the current phase-field residual whereas PIDL normally updates f_{fatigue} after training.
`history_vars(:,:,1)`	Energy-history channel in some formulations; not the damage-history variable in the current PENALTY fatigue path.	Current PENALTY path: unused, stale or zero for the relevant comparison.	No direct counterpart to PIDL `hist_alpha`.	Do not compare.	Important audit correction: compare FEM `p_field_old` with PIDL `hist_alpha` for damage history.

Cyclic Update Sequence

Figure 3.3 summarises the update sequence across loading steps. The most important difference is the timing of the fatigue update. In the FEM reference, the phase-field assembly can compute candidate values of the active driver, accumulated fatigue history and fatigue degradation during the current phase-field solve. In the PIDL implementation, the history variables are normally fixed during optimisation and refreshed only after the final field is evaluated. The new PIDL fatigue degradation factor therefore mainly affects the following step. This difference must be considered when interpreting the fatigue-life and active-driver discrepancies in Chapter 4. The storage and update audit in Table 3.5 does not invalidate the controlled benchmark, but it limits the strength of any claim that the current PIDL formulation reproduces the FEM fatigue mechanism.

The FEM–PIDL comparison requires two levels of alignment. The first is a monotonic sanity check, which tests whether the basic geometry, boundary conditions, phase-field setup and energy terms produce comparable behaviour in a non-cyclic setting. This check does not validate fatigue-history accumulation or fatigue degradation. The second is a cyclic alignment check, which verifies that the fatigue benchmark uses the same loading definition, initial crack, material parameters, phase-field length scale, initial fatigue history and update sequence before the Chapter 4 fatigue comparison is interpreted.

3.7 Chapter Summary

This chapter defined the controlled numerical setting used for the FEM–PIDL fatigue comparison. The benchmark is a two-dimensional SENT specimen with a horizontal pre-crack, normalised linear-elastic material parameters, a fixed phase-field length ratio $l_0/L = 0.01$, and displacement-controlled cyclic loading with $R = 0$. These choices define a reproducible numerical fracture problem, but they are not calibrated pavement material or loading conditions.

The chapter also specified the adopted phase-field fatigue formulation. The model combines a degraded tensile elastic-energy term, an AT1 damage-energy term, an irreversibility treatment and a fatigue degradation function controlled by the accumulated fatigue-history variable. The elastic degradation function, phase-field length scale, fatigue threshold and degradation law are treated as adopted benchmark/model choices rather than newly derived physical laws. Their role is to align the FEM and PIDL formulations within the same controlled model family.

Finally, the chapter distinguished the FEM reference from the PIDL approximation. The FEM reference stores and solves the phase-field problem through finite-element degrees

of freedom, while the PIDL formulation represents displacement and damage with neural-network fields and minimises a mesh-supported Deep Ritz energy loss. The monotonic comparison is used only as a sanity check of the basic phase-field setup. The fatigue comparison in Chapter 4 must therefore be interpreted through the cyclic update sequence and field-level diagnostics, especially the evolution of damage, fatigue history and active crack-driving fields.

Chapter 4

Current work: Comparison of the FEM–PIDL Fatigue Benchmark

4.1 Diagnostic Definitions and Comparison Procedure

The model quantities and storage conventions are defined in Chapter 3. This section defines how the Chapter 4 diagnostic quantities are calculated and interpreted before the results are presented. The purpose is to make the fatigue-life, event-timing, crack-tip and field-ratio comparisons reproducible.

The state-variable storage and update audit in Chapter 3 defines where the hidden variables are stored in each solver. A shorter result-facing version of that audit is repeated before the field-level mechanism diagnostics because those results depend directly on the active driver, fatigue-history and fatigue-degradation update semantics.

4.2 Purpose of the Chapter

Chapter 3 defined the reference finite-element method (FEM) model, the physics-informed deep learning (PIDL) formulation, and the setup steps required to compare them. This chapter presents the results of that comparison. Rather than only reporting the final numerical mismatch, it explains the diagnostic process used to make the mismatch interpretable.

The primary research question is whether the PIDL model can reproduce the crack-growth behaviour of the FEM reference when the material parameters, boundary conditions and loading sequence are known. Matching the overall fatigue life is not enough for this question. The local fracture fields must also be examined, because the model should predict not only when a crack-event indicator is reached, but also where and how damage accumulates.

Table 4.1 Diagnostic definitions used in the FEM–PIDL fatigue comparison.

Diagnostic quantity	**Definition used in this chapter**	**Level**
Fatigue life, N_f	First committed cycle at which the prescribed crack-event criterion is satisfied.	Scalar
Crack-event criterion	Fixed damage-threshold and connectivity rule used to identify whether the crack has reached the selected event condition. The same rule is applied to FEM and PIDL fields after the stated spatial mapping.	Damage field
Event cycle	First cycle satisfying the crack-event criterion. FEM and PIDL event cycles are reported separately; mapped-state comparisons also use the FEM event cycle as a reference state.	Scalar / state label
Crack-tip position	Most advanced point of the connected damaged region under the same threshold and connectivity rule.	Spatial
Crack-tip error	Distance between FEM and PIDL crack-tip positions at the same committed state or at the mapped event state.	Spatial
Process-zone region	Local region around the crack tip or high-driver zone used for field statistics. The same mask or distance rule is used before field ratios are interpreted.	Elem / field
Field ratio	PIDL value divided by FEM value for the same diagnostic field, same state label and same spatial reduction.	Scalar statistic
High-percentile field statistic	Percentile reduction, usually 99th percentile, used to compare field distributions without relying only on maxima.	Field statistic
Energy ratio	PIDL/FEM ratio for the same energy contribution, usually E_{el} or E_{d}, at a stated cycle or mapped event state.	Scalar

The first response to the mismatch was to test whether it was mainly caused by the neural-network representation. Several representation and training variants were explored during the first-year work. However, they did not remove the mismatch in the local fracture fields. This suggested that the problem should not be treated only as a network-capacity issue.

The work therefore shifted from architecture tuning to stricter FEM–PIDL comparison and field-level diagnosis. This does not mean that the alignment work was repeated from the beginning. Chapter 3 established comparison-level alignment: the two solvers were made comparable at the level of geometry, loading, material settings and initial state. Chapter 4 uses this aligned protocol to ask a different question: whether the fatigue-dependent fields evolve in the same way during cyclic loading.

The alignment checks made the comparison fair, but they did not guarantee that the fatigue-history, degradation and crack-driving fields would remain consistent. The results indicate that the remaining mismatch is mainly associated with fatigue-history accumulation, fatigue degradation and the degraded active crack-driving field. Future work should test revised history-update and crack-driving formulations to reduce the remaining field-level mismatch.

4.3 Macro-level Agreement Versus Local Field Mismatch

The work began with a scalar fatigue-life sweep. It tested whether a PIDL run could reproduce the broad relationship between displacement amplitude and cycles to failure. Figure 4.1 shows the exploratory sweep. The PIDL run follows the same monotone fatigue-life trend as the FEM reference, while the gap is larger at lower displacement amplitudes. This result suggested that the PIDL model captured a useful fatigue-life trend.

However, Figure 4.2 shows that although the early PIDL runs could appear close to the FEM reference at the scalar N_f level, the degradation histories followed different paths. The FEM reference showed stronger late local degradation, while the PIDL curves degraded more gradually over the normalised cycle range. This implies that the PIDL response was spreading fatigue degradation rather than concentrating it in the FEM crack-tip process zone.

This observation is supported by Figure 4.3. The exploratory PIDL model fails to localise damage growth and fatigue degradation along the true fracture path.

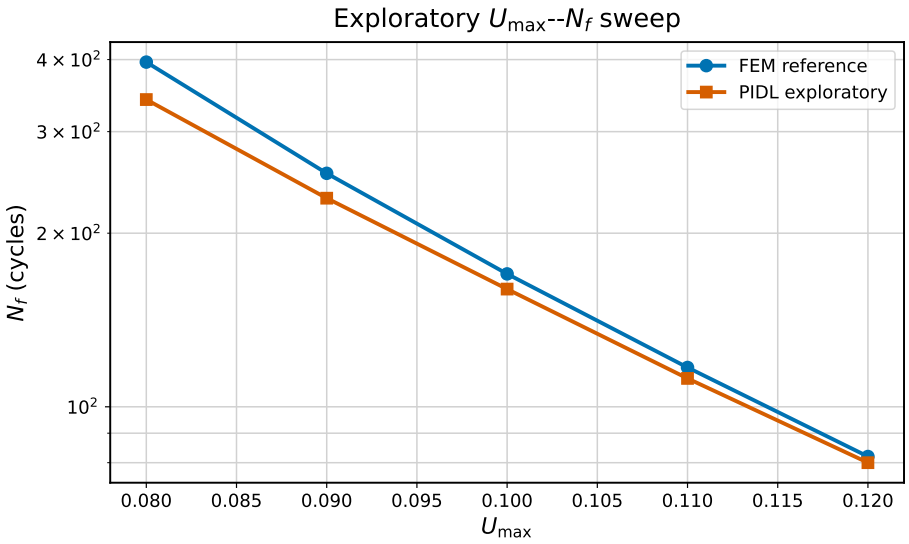


Fig. 4.1 Exploratory $U_{\max}$–N_f sweep. The PIDL run follows the same monotone fatigue-life trend as the FEM reference, while the gap is larger at lower displacement amplitudes.

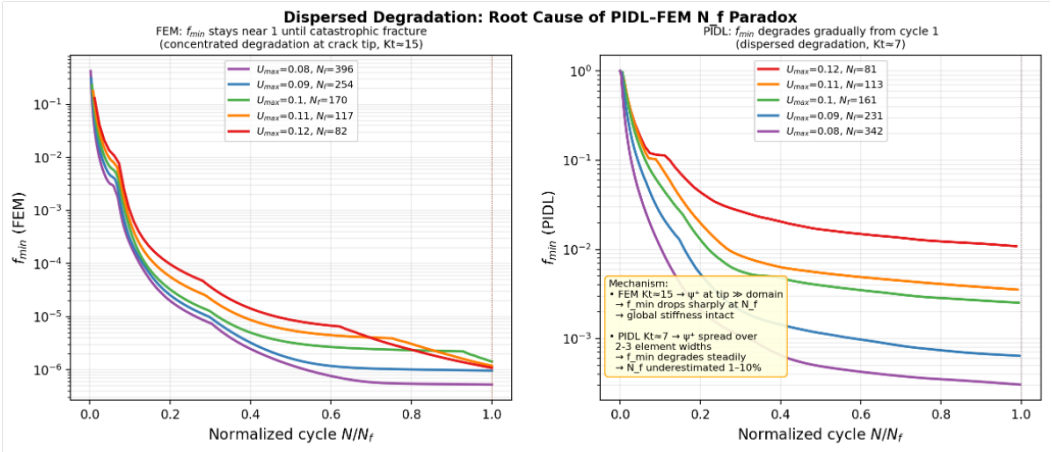


Fig. 4.2 Early comparison of minimum fatigue-degradation evolution. The figure is used as diagnostic evidence only: it explains why scalar N_f agreement was not sufficient and why local neural-network representation was investigated in the diagnostic workflow.

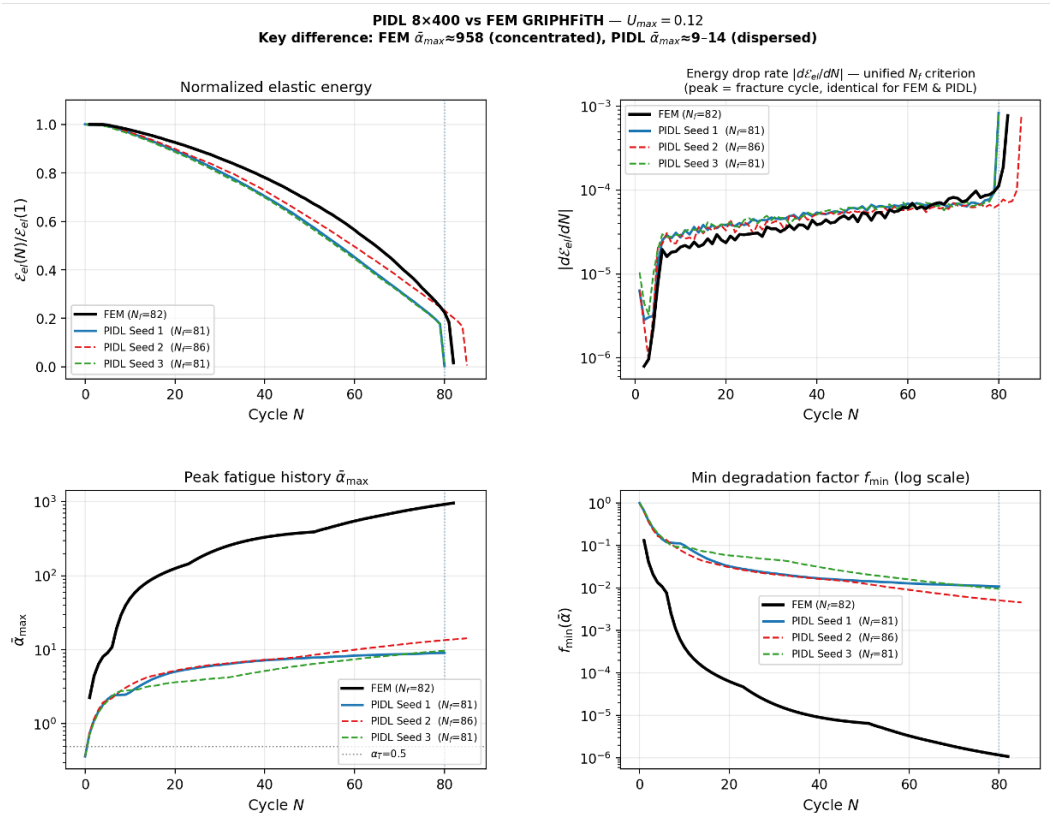


Fig. 4.3 Early single-amplitude diagnostic at $U_{\max} = 0.12$. The fracture-cycle estimates are close in this exploratory plot (FEM $N_f = 82$; PIDL seeds $N_f = 81, 86, 81$), but the internal fatigue-history and degradation quantities are not matched.

These early results imply that matching a scalar fatigue-life trend does not prove that the local fracture process is the same. Two models can reach a similar event time while accumulating damage and fatigue degradation in completely different spatial patterns. This matters for the longer-term inverse problem because inspection-informed prediction will need local fracture-state fields, not only a single N_f value.

4.4 First Diagnostic Branch: Attempts to Reduce the Field Mismatch

Field-level mismatch motivated the first branch of the investigation: whether the mismatch was mainly caused by the PIDL representation. This hypothesis was motivated by spectral bias in coordinate-based neural networks, where highly localised fields can be harder to represent than smooth fields [45]. The concern was that the PIDL network might smooth the local crack-tip driving field rather than reproduce the sharper FEM process zone.

Several exploratory methods were tested to enhance the PIDL representation. These included Williams-style crack-tip features, Fourier features, local correction heads, crack-tip patches, overlapping local subnetworks, discontinuity-aware inputs and sampling or history-weight changes. Table 4.2 summarises part trials of this branch. Further details on the tested methods are provided in Appendix A.1.

However, the results were not ideal. Figure 4.4 shows several variants gave scalar event times close to the FEM reference, but the peak fatigue-history value remained far below the FEM scale. These runs are useful as early diagnostics, but they are not used as final evidence because they changed event timing or local patterns without reliably removing the field mismatch. Therefore, the mismatch should not be treated only as a network-capacity problem. The next question was whether the remaining field mismatch could be caused by comparison-level issues rather than the PIDL formulation itself.

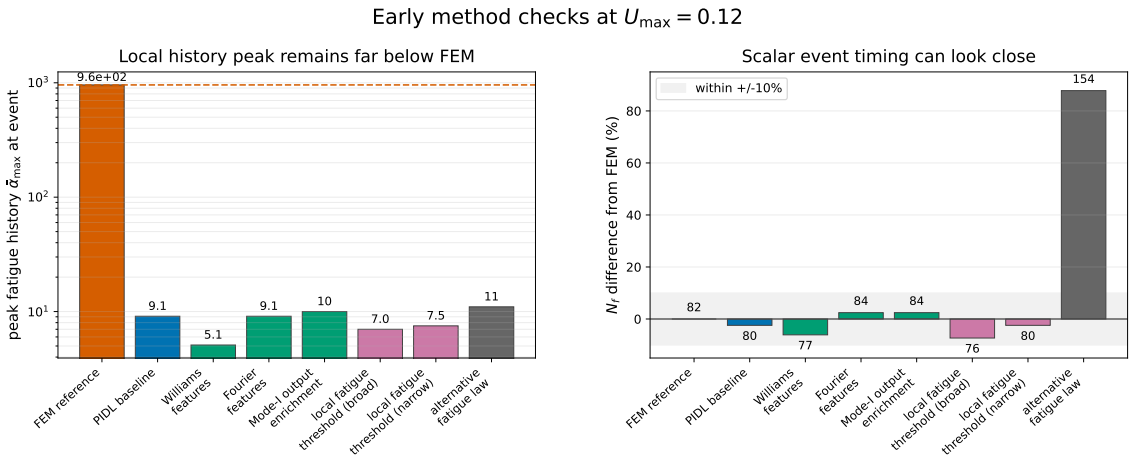


Fig. 4.4 Summary of early method checks at $U_{\max} = 0.12$. The left panel compares the peak fatigue-history value at each method's event, and the right panel compares the event cycle against the FEM reference. The comparison shows that several exploratory PIDL variants gave event cycles near the FEM value, but the local fatigue-history peak remained much lower than the FEM reference.

Table 4.2 Method map from early exploratory trials to fix the field mismatch.

Hypothesis	Change tested	Observed result	What it taught us	Next step
Neural-network expressivity	Williams-style crack-tip features, Fourier features and Mode-I output enrichment.	Some event cycles stayed close to FEM, but the fatigue-history peak did not reach the FEM scale consistently.	Extra local features did not by themselves force FEM-like crack-tip localisation.	Check setup alignment.
Local representation and sampling	Local patches, overlapping subnetworks, discontinuity-aware inputs and process-zone weighting.	Local patterns or event timing changed, but no robust field-level match was retained.	Increasing local flexibility can change the optimisation path without proving comparability.	Use mapped field metrics.
Fatigue-chain modification	Spatial fatigue-threshold variants, alternative accumulation laws and history weighting.	Kinetics and event timing changed, but the process-zone history/degradation mismatch remained.	Downstream fatigue changes can compensate scalar timing without fixing the driver field.	Separate raw and active drivers.

4.5 Second Diagnostic Branch: Fatigue Benchmark Diagnosis

The previous section showed that changing the PIDL architecture or adding local features could alter event timing and local field patterns, but failed to reliably remove the field mismatch. This implies that the bottleneck is not simply a matter of network capacity, but is probably rooted in how the fields change over time. Therefore, this section tried to track the continuous evolution trajectories of the field variables. We investigate whether the coupled

fracture and fatigue fields in the PIDL framework evolve in the same way as the FEM reference and make the field disagreements physically interpretable.

During this exploration, we found that the fatigue comparison required extra care because cyclic fracture evolution depends heavily on state history. Small differences in boundary conditions or initial states can change the stress-localisation patterns near the pre-crack, altering where fatigue history accumulates during loading. So the initial state was checked first.

4.5.1 Initial-State Verification

The comparison uses two solvers of the same controlled SENT specimen under repeated loading with a displacement amplitude of 0.12. The FEM simulation provides the controlled phase-field fracture reference. The PIDL model is trained and evaluated on the corresponding setting.

It is noted that the initial-crack treatment was revised to make the FEM and PIDL settings more directly comparable. In the previous FEM setup, the pre-crack region was initialised as a fully damaged band while the fatigue history was set to zero, and the model was then relaxed under zero displacement before cyclic loading. Here, the FEM reference uses the same smooth diffuse initial-crack profile as the PIDL setting, with zero initial fatigue history, and then proceeds directly to the prescribed cyclic loading sequence.

The initial-state comparison confirms that the starting fields are effectively matched. The maximum damage values in FEM and PIDL are both approximately 0.905, and the mean damage values are both approximately 0.0378. The fatigue-history variable is zero in both models, and the fatigue-degradation function f_{fatigue} is initially equal to one.

need a residual figure for initial state here

This initial agreement does not guarantee field-level agreement during fatigue loading. It only shows that the later divergence is not caused by an obvious mismatch in the initial damage field, fatigue-history field or degradation state. The later comparison can therefore be interpreted as a fatigue-evolution problem rather than as an initial-state artefact.

4.5.2 Global Scalar Divergence and Event-Timing Lag

The scalar comparison was examined in the initial-state verification. These scalar quantities test whether the PIDL run follows the broad fatigue progression of the FEM reference before the local fields are interpreted.

The FEM reference reaches the crack-event criterion at cycle 69. The PIDL run detects the corresponding event later than the FEM reference. This timing difference is not so

large that the PIDL response is unrelated to the FEM response, but it shows that the cyclic propagation process is delayed in the PIDL formulation.

The early energy response is close. At the first peak loading state, the PIDL-to-FEM elastic-energy ratio is approximately 0.991, and the corresponding crack-surface-energy ratio is approximately 0.955. These values support the interpretation that the early loading response and energy scale are not obviously inconsistent.

The late-cycle energy response is different. By the FEM crack-event cycle, the PIDL-to-FEM elastic-energy ratio increases to approximately 4.789, while the crack-surface-energy ratio falls to approximately 0.448. The PIDL run therefore carries too much elastic energy and accumulates too little crack-surface energy relative to the FEM reference near the late fracture stage.

Table 4.3 Selected scalar indicators from the FEM–PIDL fatigue comparison.

Quantity	Early-cycle comparison	Late-cycle comparison
Damage maximum	FEM/PIDL both approximately 0.905 initially	Similar order, but not sufficient to indicate field agreement
Damage mean	FEM/PIDL both approximately 0.0378 initially	Comparable global mean, but local differences remain
Elastic energy	PIDL/FEM ratio 0.991 at first peak	PIDL/FEM ratio 4.789 at the FEM event cycle
Crack-surface energy	PIDL/FEM ratio 0.955 at first peak	PIDL/FEM ratio 0.448 at the FEM event cycle
Event timing	Not applicable at initial state	FEM reaches event at cycle 69; PIDL event occurs later

4.5.3 Field-Level Mechanism Diagnostics

The field-level comparison shows that the scalar divergence is associated with fatigue-history and degradation mismatch near the process zone. At the FEM event cycle, the maximum fatigue-history value in the FEM reference is approximately 19.53, while the corresponding PIDL value is approximately 2.00. The minimum fatigue-degradation value is approximately 0.0146 in FEM and 0.170 in PIDL. Since smaller values of f_{fatigue} indicate stronger local fatigue weakening, the PIDL formulation does not accumulate or degrade the local fatigue state as strongly as the FEM reference.

The excessive late-cycle elastic energy is consistent with an under-developed local damage process zone in the PIDL solution. In the FEM reference, fatigue-history accumulation near the crack tip reduces f_{fatigue}, lowers the effective damage-energy resistance and allows stronger local damage growth. This reduces the local stiffness and increases the

Table 4.4 Result-facing audit of fatigue-history and driver quantities used in Chapter 4.

Quantity	**FEM treatment**	**PIDL treatment**	**Diagnostic implication**
Damage α	Nodal phase-field damage `p_field`; interpolated to GP during assembly and reduced for field comparison.	Neural damage output `alpha`; evaluated at nodes and reduced to `alpha_elem` for element diagnostics.	Defines the current crack state and process-zone map.
Damage history	Committed nodal maximum damage `p_field_old`.	Committed nodal maximum damage `hist_alpha`.	Correct pair for irreversibility-state comparison. FEM `history_vars(:,:,1)` is not used as PIDL damage history.
Raw driver ψ_{raw}	Undegraded tensile driver `strain_en_undgr`; GP quantity reduced to element level.	`psi_raw_elem`; elementwise diagnostic.	Similar raw-driver maxima do not prove fatigue-mechanism agreement.
Active driver ψ_{active}	$g(d_q)Y_q$ at GP level, reduced for comparison.	`psi_active_elem` or probe-reduced element diagnostic.	Central quantity for fatigue-history accumulation.
Accumulated fatigue history $\bar{\alpha}$	`history_vars(:,:,2)`; GP quantity reduced to element level.	`hist_fat` or `alpha_bar`; elementwise history.	Main underpredicted field in the PIDL result.
Fatigue degradation f_{fatigue}	`history_vars(:,:,4)`; GP quantity reduced to element level.	`f_fatigue`; elementwise degradation factor.	Controls effective damage-energy resistance.
Current fatigue increment	Positive increment from the current active driver relative to previous GP baseline.	Use `delta_alpha_bar_input` if saved, `psi_plus_prev_elem` may already be the next-step baseline.	Avoids misreading the saved baseline as the current increment.

damage-energy contribution associated with crack-surface formation. In the PIDL result, fatigue-history accumulation and fatigue degradation remain weaker near the corresponding process zone. Under the same imposed displacement, the PIDL solution therefore retains a larger part of the response in the elastic-energy term and develops a smaller damage-energy contribution. This interpretation is supported by the late-cycle ratios $E_{\mathrm{el}}^{\mathrm{PIDL}}/E_{\mathrm{el}}^{\mathrm{FEM}} \approx 4.789$ and $E_{\mathrm{d}}^{\mathrm{PIDL}}/E_{\mathrm{d}}^{\mathrm{FEM}} \approx 0.448$, together with the underprediction of $\bar{\alpha}$ and weaker fatigue degradation.

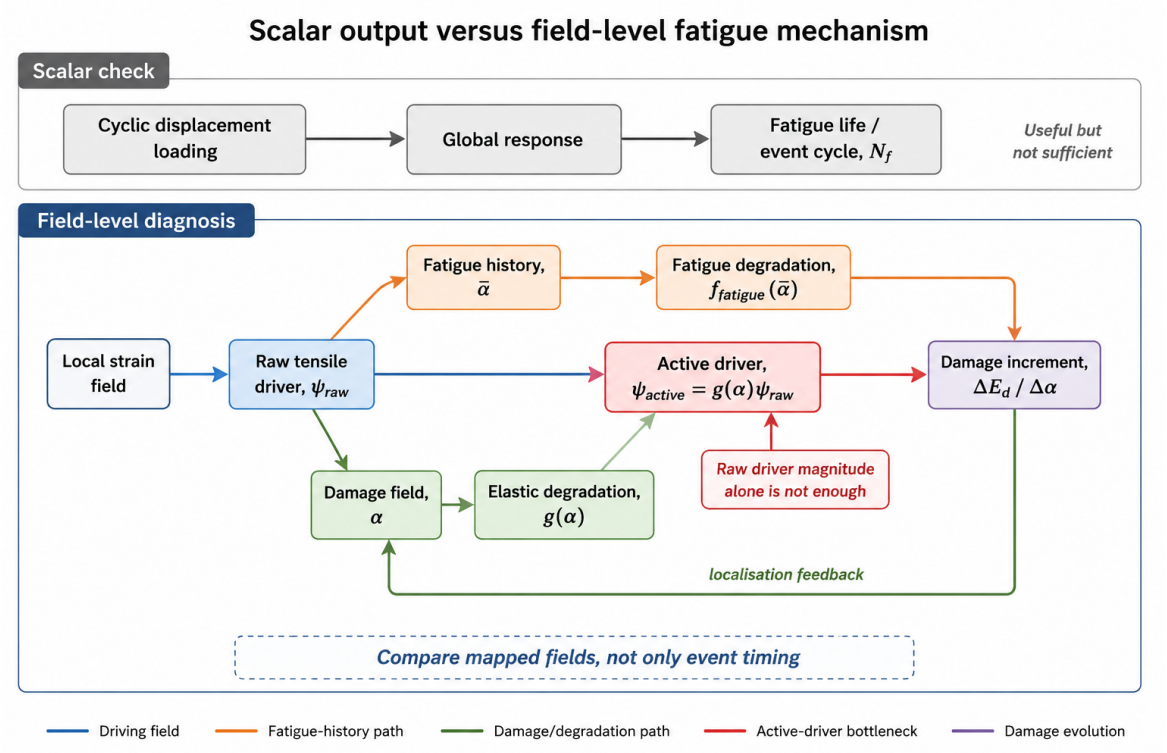


Fig. 4.5 Coupled phase-field fatigue variables used in the FEM–PIDL comparison. The raw tensile driver ψ_{raw}, damage field α, fatigue history $\bar{\alpha}$, elastic degradation $g(\alpha)$, active driver $\psi_{\mathrm{active}} = g(\alpha)\psi_{\mathrm{raw}}$ and damage-energy increment ΔE_{d} are used to interpret the field-level mismatch.

The driver diagnostics test whether this mismatch is caused by insufficient raw tensile driving energy. At the FEM event cycle, the maximum values of the raw tensile driver ψ_{raw} approximately 2433.7 in FEM and 2638.7 in PIDL. However, the high-percentile distribution is very different. The 99th percentile of ψ_{raw} is approximately 500.1 in FEM, but only approximately 0.555 in PIDL. Thus, although the PIDL result contains a large local raw-driver maximum, it does not form the same high-driver process-zone distribution as the FEM reference.

The active driver shows the mismatch more directly. At the FEM event cycle, the 99th percentile of $\psi_{\text{active}} = g(\alpha)\psi_{\text{raw}}$ is approximately 0.1059 in FEM and 0.0336 in PIDL. Region-based reductions give the same result. In the pre-crack band, the PIDL-to-FEM raw-driver ratio is approximately 2.45, but the active-driver ratio is only approximately 0.008. In the tip-column region, the raw-driver ratio is approximately 1.18, while the active-driver ratio is approximately 0.096.

Table 4.5 Raw and active crack-driving quantities at the FEM event cycle.

Region and quantity	FEM	PIDL	PIDL/FEM
Pre-crack band, raw	72.31	176.97	2.45
Pre-crack band, active	2.09×10^{-2}	1.66×10^{-4}	0.008
Tip column, raw	26.58	31.39	1.18
Tip column, active	3.06×10^{-3}	2.95×10^{-4}	0.096

These reductions show that the bottleneck is not the raw tensile-energy magnitude alone. The PIDL formulation can generate large local values of ψ_{raw}, but these values do not become a FEM-like active driver in the process zone.

A raw-driver diagnostic test was used to check this interpretation. The test replaced ψ_{active} with ψ_{raw} in the history update to examine whether the late process-zone lag could be removed by using a stronger driving signal. This was a diagnostic test only, not a proposed physical correction. It moved the first detected event one cycle earlier and reduced the mean absolute crack-tip error from approximately 0.0403 to 0.0323. However, it did not recover the FEM event-cycle process zone: at the FEM event-cycle mapping, the PIDL high-damage crack-tip position remained below the FEM value, approximately 0.871 compared with 0.999.

The remaining mismatch is therefore best interpreted as a coupled fatigue–fracture feedback problem. In the FEM reference, local strain concentration, active crack-driving field, fatigue-history accumulation, fatigue degradation and stiffness redistribution are updated together through the phase-field solution process. In the current PIDL formulation, this loop is only approximated through the physics-informed loss and the history update used during training. Future work should therefore target the fatigue-history update, degradation response and active crack-driving formulation, rather than focusing only on architecture tuning or scalar event timing.

4.6 Chapter Summary

This chapter examined why scalar fatigue-life agreement was not sufficient for the FEM–PIDL fatigue benchmark. The exploratory $U_{\max}$–N_f sweep showed that the PIDL formulation could follow the broad monotone fatigue-life trend of the FEM reference. However, the accompanying local fields showed that similar event timing could still be produced with different fatigue-history and degradation patterns. The first diagnostic branch therefore tested whether the mismatch was mainly caused by neural-network representation. Williams-style features, Fourier features, local correction terms, local patches, overlapping subnetworks and history-weighting variants could change event timing or local field appearance, but they did not reliably recover a FEM-like fatigue-history peak or process-zone field.

The second diagnostic branch used the aligned protocol from Chapter 3 to make the cyclic fatigue comparison interpretable. The FEM reference and PIDL setting were compared on the same controlled SENT benchmark, with the same smooth diffuse initial-crack profile and zero initial fatigue history. The initial fields were effectively matched: the maximum damage values were both approximately 0.905, the mean damage values were both approximately 0.0378, $\bar{\alpha} = 0$ in both models, and $f_{\text{fatigue}} = 1$ initially. This means that the later mismatch is not best explained by an obvious initial-state error.

The scalar and field-level results then showed where the divergence developed. At the first peak loading state, the PIDL-to-FEM elastic-energy and crack-surface-energy ratios were close to one, approximately 0.991 and 0.955, respectively. By the FEM crack-event cycle, cycle 69, the PIDL event was delayed, the elastic-energy ratio increased to approximately 4.789, and the crack-surface-energy ratio fell to approximately 0.448. The corresponding local fields support an under-developed-process-zone interpretation: the FEM fatigue-history maximum was approximately 19.53, compared with approximately 2.00 in PIDL, and the minimum fatigue-degradation value was approximately 0.0146 in FEM, compared with approximately 0.170 in PIDL. The excessive late-cycle elastic energy is therefore consistent with the PIDL formulation not accumulating and degrading the local fatigue state strongly enough near the process zone.

The crack-driving diagnostics showed that this mismatch is not solved by considering the raw tensile driver alone. At the FEM event cycle, the maximum ψ_{raw} values were of the same order in FEM and PIDL, approximately 2433.7 and 2638.7, but the 99th percentile values were very different, approximately 500.1 in FEM and 0.555 in PIDL. The active driver showed the limitation more directly: the 99th percentile of ψ_{active} was approximately 0.1059 in FEM and 0.0336 in PIDL, and the regional active-driver ratios were much smaller than the corresponding raw-driver ratios. A raw-driver diagnostic test improved some scalar indicators, reducing the mean absolute crack-tip error from approximately 0.0403 to 0.0323,

but it did not recover the FEM event-cycle process zone. The main conclusion is therefore that the current PIDL formulation can match the initial state and part of the early scalar response, but it does not yet reproduce the coupled feedback between active crack-driving field, fatigue-history accumulation, fatigue degradation and stiffness redistribution. Future work should target this coupled fatigue-history and active-driver formulation, and should continue to evaluate it using mapped field-level diagnostics as well as scalar event indicators.

Chapter 5

Research Proposal and Future Work

5.1 Research Direction

Chapters 1 and 2 established the practical motivation for this research. Pavement inspection can provide visible crack and condition information, including crack masks, crack geometry, crack width, surface distress indicators and repeated survey records. However, these inspection outputs do not directly provide the hidden fracture-state variables that control future local crack growth. In particular, they do not directly reveal the internal damage field, fatigue-history field, active crack-driving field, local material degradation or the loading history experienced by the pavement. Chapters 3 and 4 then tested a stricter and more controlled version of this problem. A phase-field finite-element model was used as the reference solution, and a physics-informed deep learning (PIDL) solver was tested under aligned geometry, material, boundary-condition and cyclic-loading settings. The purpose of this first-year modelling stage was not to validate a field-scale pavement predictor. It was to ask a diagnostic question: if the geometry, material parameters and loading family are already known, can a physics-only neural solver reproduce the hidden fracture-state evolution without FEM or data labels?

The main lesson from Chapter 4 is that physics-only supervision is not sufficient in the current formulation. The PIDL model can reproduce the initial state and some scalar trends, but scalar agreement does not guarantee reliable hidden-field prediction. A model may show a partly plausible crack-tip position or crack-growth timing while still producing the wrong fatigue-history accumulation, fatigue-degradation field, active crack-driving field or stiffness redistribution. This matters because future crack growth depends on these local fields rather than on a scalar crack indicator alone.

This result changes the role of the future work. The future programme will not treat pure PIDL as the main prediction route. Instead, the first-year PIDL study is used as diagnostic

evidence for the importance of data supervision and field-level validation. Once trusted simulation data, partial inspection observations or structural proxy data are available, the research can move from physics-only diagnosis to data-supported identification and data-supervised prediction. In this chapter, *identification* means estimating the hidden fracture state, initial condition, material parameters or loading proxies from partial observations. *Prediction* means propagating the identified state forward to estimate future damage or crack evolution. The central future direction is therefore:

> The first-year PIDL study shows the limit of physics-only supervision for fatigue crack-growth prediction. Even under controlled and known benchmark conditions, scalar agreement does not guarantee reliable hidden-field prediction. Future work therefore needs data: first to identify hidden fracture states and uncertain model inputs from partial observations, and then to predict future crack evolution with supervised or hybrid models.

The future work is organised into three work packages. Work Package 1 starts with data-supported identification of crack-growth initial states and uncertain inputs. It asks which hidden state variables can be inferred from crack masks, crack width, repeated observations, structural proxies and synthetic FEM degradation tests. Work Package 2 develops supervised crack-growth prediction models, especially uncertainty-calibrated neural operators, using trusted phase-field trajectories and identified states as inputs. Work Package 3 validates the full chain under partial data: partial observation, state/input identification, prediction rollout, and comparison against known FEM fields or repeated observations. The three work packages are sequential and cumulative rather than competing alternatives. WP1 asks what state can be identified. WP2 asks how the identified state can be propagated. WP3 asks whether the identification–prediction chain remains reliable when observations are partial, noisy and non-unique. Physics-informed terms from the PIDL study are retained, but their role is changed. They are no longer the only source of supervision. They become diagnostic constraints, ablation variables, validation gates and optional regularisers inside a data-supervised workflow.

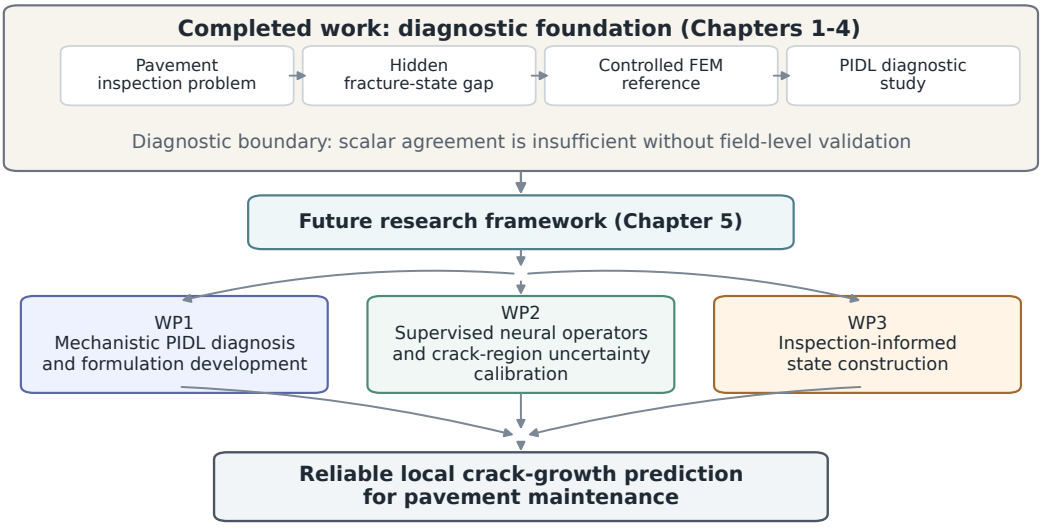


Fig. 5.1 Future research roadmap. Chapter 5 proposes a data-centred route from inspection outputs and partial observations, to hidden-state and uncertain-input identification, to supervised crack-growth prediction, and finally to hybrid validation under partial data. The Chapter 4 PIDL diagnosis is retained as evidence that physics-only supervision is insufficient and that field-level validation is necessary.

5.2 Proposed Research Work Packages

5.2.1 Work Package 1: Data-supported Identification of Crack-growth Initial States and Uncertain Inputs

Work Package 1 (WP1) replaces the original plan to continue pure PIDL as the first future work package. This change follows directly from Chapters 3 and 4. The completed PIDL benchmark showed that, even under controlled and known settings, physics supervision alone did not reliably recover the hidden local fields needed for crack-growth prediction. The next step is therefore not to keep tuning the same pure physics-supervised solver as the main research route. The next step is to ask what can be identified once observations and data are introduced. WP1 treats inspection and simulation-derived observations as partial evidence about an unobserved fracture state. The observable data may include crack masks, crack skeletons, crack width, crack-tip locations, repeated images, survey timestamps, surface-condition descriptors, structural response proxies and traffic or loading descriptors. These observations are denoted abstractly as

$$\mathbf{y}_t = \left[M_t(\mathbf{x}),\, w_t(\mathbf{x}),\, \mathbf{x}_{\mathrm{tip},t},\, s_t(\mathbf{x}),\, p_{\mathrm{load},t}\right],$$

where M_t is the registered crack mask, w_t is the crack-width observation where available, $\mathbf{x}_{\mathrm{tip},t}$ denotes extracted crack-tip information, s_t represents structural or surface-condition proxies, and $p_{\mathrm{load},t}$ denotes traffic or loading descriptors. The hidden variables of interest are not directly observed images. They are fracture-state and model-input quantities such as

$$\mathbf{z}_t = \left[\alpha_t(\mathbf{x}),\, \bar{\alpha}_t(\mathbf{x}),\, \psi_{\mathrm{active},t}(\mathbf{x}),\, f_{\mathrm{fatigue},t}(\mathbf{x}),\, \theta_{\mathrm{mat}},\, \theta_{\mathrm{load}}\right],$$

where α_t is the phase-field damage, $\bar{\alpha}_t$ is the accumulated fatigue-history field, $\psi_{\mathrm{active},t}$ is the active crack-driving field, $f_{\mathrm{fatigue},t}$ is the fatigue-degradation field, and θ_{mat} and θ_{load} represent material and loading parameters. The purpose of WP1 is not to claim that this full state can be uniquely recovered from a crack image. The purpose is to measure which components are identifiable, which components remain uncertain, and which additional data reduce uncertainty.

WP1.1: Registration and observation alignment. The first task is to align crack observations from different images, survey times or camera positions into a common pavement coordinate system. Without alignment, apparent crack growth may be caused by camera motion, perspective change, scale variation or inconsistent survey positioning rather

than physical crack extension. If the road surface can be approximated as a local plane, homography-based registration will be used. For video or repeated survey sequences, a dynamic homography can be estimated over time. If camera calibration, depth information, control points or reconstructed three-dimensional points are available, a Perspective-n-Point (PnP), RGB-D or SfM-assisted route can be tested. The output will be registered crack masks $M_t(\mathbf{x})$, crack-width observations $w_t(\mathbf{x})$ where available, and registration uncertainty. Registration will be treated as an uncertainty source rather than perfect preprocessing. Registration uncertainty can be quantified through reprojection error, feature-matching residuals, repeated-mask overlap or crack-tip localisation error. This uncertainty will be propagated into later identification and prediction stages.

WP1.2: Crack geometry extraction and proxy-state construction. The second task is to convert registered crack observations into model-relevant quantities. A raw crack mask is not a fracture-state field. From each registered crack mask, geometric descriptors will be extracted, including crack skeleton, endpoints, branch points, connected components, local orientation, crack length, tortuosity and local width. These descriptors define the observable part of the crack state. An initial damage proxy can be constructed from the distance to the observed crack set. If $D(\mathbf{x}, M_t)$ is the distance from point $\mathbf{x}$ to the registered crack mask M_t, a simple initial damage proxy may be written as

$$\alpha_0(\mathbf{x}) = \exp\left[-\frac{D(\mathbf{x}, M_t)}{\ell}\right],$$

or as a bounded sigmoid profile controlled by crack width and phase-field length scale. This construction does not claim to recover the true internal damage field. It provides a reproducible initial condition that can be varied during identification. Additional auxiliary channels can be added to represent physically meaningful but uncertain information. A crack-tip indicator $T(\mathbf{x})$ can be generated from endpoints and branch tips. A low-cost crack-driving proxy $S(\mathbf{x})$ can be computed using distance-and-angle fields around crack tips or simplified fracture-mechanics priors. Material heterogeneity can be represented through a latent random field $m(\mathbf{x})$ or a small set of material descriptors. Loading uncertainty can be represented through descriptors p_{load}. These quantities form a distribution of plausible initial states rather than a single deterministic reconstruction.

WP1.3: Synthetic and semi-synthetic identification tests. Before applying the identification pipeline to real inspection records, it will be tested using controlled FEM or phase-field fields. Full hidden fields will be degraded into inspection-like observations such as binary

crack masks, sparse crack-tip observations, noisy crack-width maps, partial damage fields or surface-only measurements. Because the true hidden state remains known in these tests, the identification error can be measured directly. The synthetic identification tests will compare several observation levels:

1. crack mask only;
2. crack mask plus crack width or surface-profile information;
3. repeated crack observations over time;
4. crack observations plus structural response proxies;
5. crack observations plus approximate traffic or loading descriptors.

This comparison is central to the thesis argument. It tests how much data are needed before hidden fracture-state variables and uncertain model inputs become identifiable enough for prediction.

WP1.4: Initial-state and uncertain-input inference. The fourth task is to infer the uncertain current state and model inputs from partial observations. Several routes can be compared. A direct encoder can map observation channels to a latent state distribution. An optimisation-based inverse method can minimise mismatch between observed crack geometry and predicted or reconstructed state variables. An ensemble or Bayesian approximation can be used to represent non-uniqueness. The output should not be a single overconfident state. It should be an uncertainty-aware estimate of α_t, fatigue-history proxies, material parameters and loading descriptors. The inverse problem can be written abstractly as

$$\hat{p}(\mathbf{z}_t, \theta \mid \mathbf{y}_t) \quad \text{or} \quad (\hat{\mathbf{z}}_t, \hat{\theta}) = \arg\min_{\mathbf{z}_t, \theta} \mathscr{L}_{\text{obs}}(\mathbf{y}_t, \mathscr{H}(\mathbf{z}_t, \theta)) + \mathscr{R}(\mathbf{z}_t, \theta),$$

where $\mathscr{H}$ is an observation operator from hidden state to observable crack or structural quantities, and $\mathscr{R}$ includes regularisation or weak physical constraints. The main output of WP1 is therefore not simply a registered image. It is an identified and uncertainty-aware crack-growth initial condition.

The deliverable of WP1 is a data-supported identification pipeline. It will convert partial observations into uncertain initial states and uncertain model inputs for crack-growth prediction. It will also define what cannot be identified from inspection alone.

Table 5.1 Identification questions in Work Package 1.

Available evidence	Identification target	Expected test
Crack mask and geometry	Initial damage proxy, crack-tip location and crack morphology	Compare reconstructed damage support, crack-tip error and uncertainty against known synthetic fields
Crack width or surface profile	Damage-zone width, process-zone scale and severity proxy	Test whether width reduces ambiguity in α_0 and ℓ
Repeated observations	Direction of active growth and accumulated change	Test whether temporal observations reduce fatigue-history and loading uncertainty
Structural or condition proxies	Material/loading descriptors and degradation indicators	Test whether proxy channels improve prediction beyond crack geometry alone
Synthetic FEM degradation	Known hidden damage, history and active-driver fields	Quantify identifiability limits and state-estimation uncertainty directly

5.2.2 Work Package 2: Supervised Crack-growth Prediction

Work Package 2 (WP2) develops the prediction route. It follows directly from the Chapter 4 diagnosis: field-level crack evolution is difficult to reproduce reliably with physics-only supervision. A supervised neural operator offers a complementary route because it learns the solution operator from trusted phase-field FEM trajectories rather than attempting to discover the solution only through a physics loss. The first task is a controlled state-to-state prediction problem. The initial target is

$$\alpha(\cdot,t) \mapsto \alpha(\cdot,t+1),$$

where the input is the current phase-field damage state and the output is the next damage field with an associated uncertainty estimate. This is a damage-field surrogate, not yet a full mechanics surrogate. Later versions will include loading descriptors, material descriptors, fatigue-history variables and active crack-driving fields where they are available. The prediction route will eventually take identified states from WP1 as inputs. The immediate supervised data source for this starting point is the PFM-Fracture benchmark [23], released through Harvard Dataverse (DOI: 10.7910/DVN/KZDRUE). This dataset is used as a separate supervised surrogate benchmark rather than as the controlled cyclic-fatigue FEM reference studied in Chapters 3 and 4. The current supervised study used the shear-spectral subset, consisting of 1,000 HDF5 simulation files on a 128×128 grid over 101 time levels. In the baseline processing, the damage channel was extracted from each file and consecutive time levels were converted into one-step pairs $\alpha(\cdot,t) \mapsto \alpha(\cdot,t+1)$, giving 100,000 input–output samples. The split was made at file level, not sample level, with 700 trajectories for training, 150 for validation and 150 for testing, so that time steps from one simulation could not leak across train and test sets.

Table 5.2 Planned supervised operator-learning task in Work Package 2.

Stage	Inputs	Outputs	Reliability audit
Initial supervised benchmark	Current damage field $\alpha(\cdot,t)$ under a fixed benchmark family	Next damage field $\alpha(\cdot,t+1)$ and ensemble uncertainty	Global error, crack-region coverage, local calibration and rollout stability
Fatigue-relevant extension	Damage, cycle or loading descriptors, fatigue-history information and material parameters where available	Damage, fatigue-history and active crack-driving fields over one or more future steps	Field-level morphology, history consistency, damage bounds and irreversibility
Identification-informed extension	Identified state from WP1, crack geometry, partial observations and structural or loading proxies	Multi-step crack-growth prediction with propagated state uncertainty	Crack-region reliability, state uncertainty and consistency with Chapter 4 validation gates

This work package already has a concrete starting point in a parallel calibration-aware neural-operator study. The study compared FNO, CA-FNO, U-FNO, FNO-wide, CNO and U-Net configurations. Its main result was a reliability result rather than only a model-ranking result. The standard FNO ensemble was overconfident near cracks, with only 78% crack-region coverage at a 95% nominal level. The CA-FNO variant adds a local spatial branch to the FNO layer and changed the crack-region coverage gap from $+0.112$ to -0.038. This supports the need for crack-region calibration before fast surrogates are used for damage-critical prediction.

WP2 will not treat low global error as sufficient evidence of reliability. Future supervised models will be evaluated using local crack-region metrics, multi-step rollout behaviour, field-level consistency and uncertainty calibration. The first extension will be to fatigue-relevant trajectories and controlled loading variations. Later extensions will add pavement-relevant material ranges and identified initial states from WP1. The deliverable of WP2 is a calibrated supervised crack-growth surrogate. It should predict future damage or crack-state fields from known or identified current states, and it should quantify uncertainty locally near crack tips and active damage zones.

5.2.3 Work Package 3: Hybrid Validation: Identification and Prediction under Partial Data

Work Package 3 (WP3) validates the complete chain under partial data. This work package is not just a registration pipeline, and it is not a return to pure PIDL. Its purpose is to test

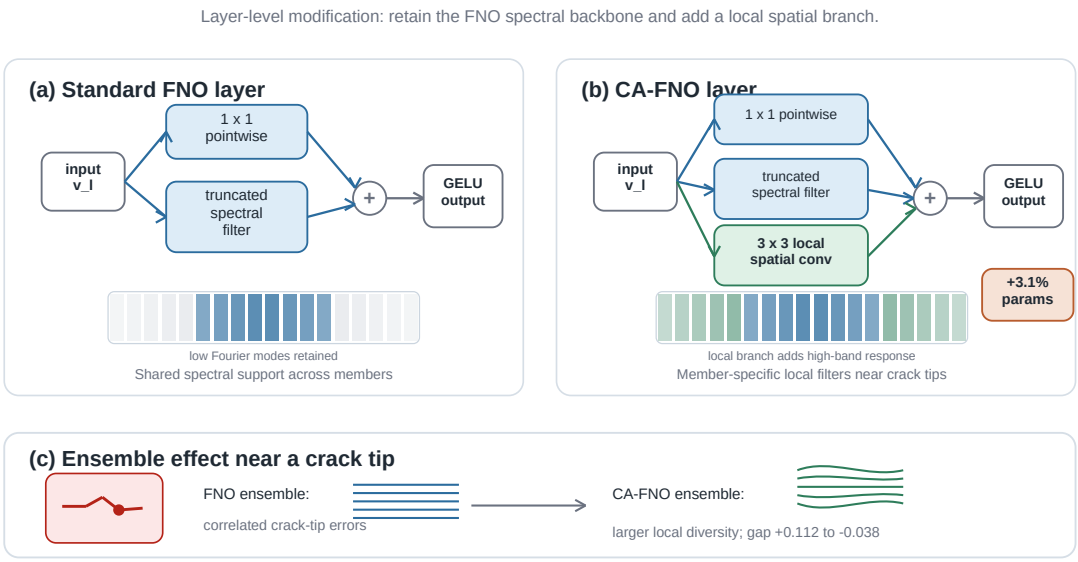


Fig. 5.2 CA-FNO architecture used in the supervised surrogate exploration. The standard FNO layer combines pointwise and truncated spectral branches. CA-FNO adds a parallel 3×3 local spatial convolution branch, with the aim of increasing member-specific diversity near crack tips where standard FNO ensembles can be overconfident.

whether partial observations can be converted into useful state and input estimates, whether those estimates improve future prediction, and whether physics-informed constraints improve the chain once data supervision is available. The core validation loop is

$$\mathbf{y}_t \;\rightarrow\; \hat{\mathbf{z}}_t, \hat{\theta} \;\rightarrow\; \hat{\mathbf{z}}_{t+\Delta t} \;\rightarrow\; \text{field-level and uncertainty audit.}$$

The first arrow is identification, the second arrow is prediction, and the final stage is validation. In synthetic or semi-synthetic tests, the hidden FEM fields are known, so both identification error and prediction error can be measured. In repeated inspection tests, the later observation provides a weaker but practically relevant validation target.

WP3.1: Partial-observation validation benchmarks. The first task is to construct validation benchmarks in which the hidden state is known but the model is only given degraded observations. Full FEM or phase-field trajectories will be converted into partial observations: crack masks, crack-tip points, noisy width maps, missing regions, sparse surface samples or repeated observations at selected times. These degraded observations will be designed to mimic what inspection can and cannot provide. This benchmark will separate three questions that are often mixed together. First, can the current hidden state be identified from the

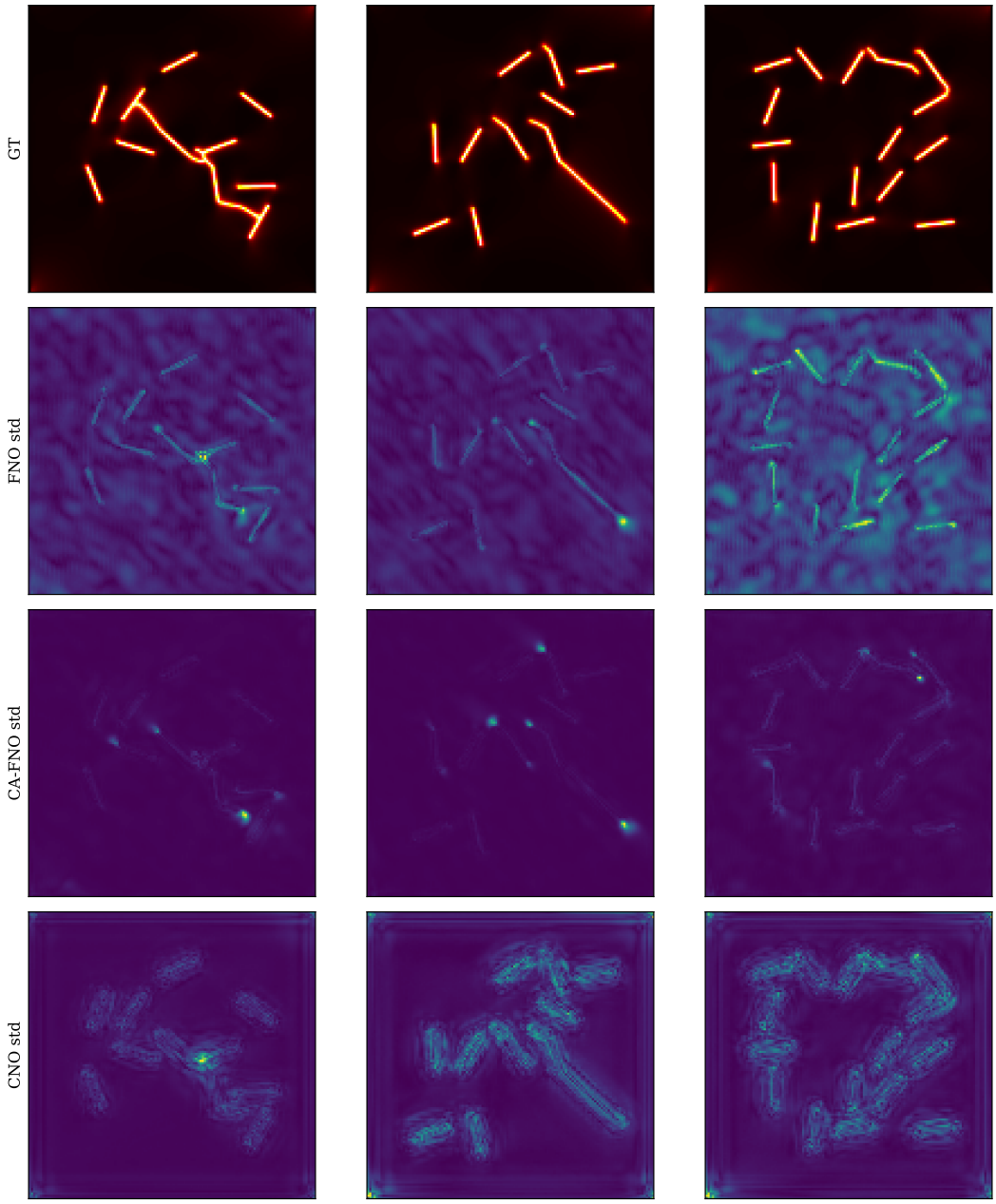


Fig. 5.3 Preliminary phase-field uncertainty fields from the supervised surrogate study. The top row shows ground-truth phase-field damage fields for held-out samples, while the lower rows show predictive uncertainty fields for FNO, CA-FNO and CNO ensembles. The figure illustrates that the supervised route has already been tested on spatial phase-field outputs and that reliability must be assessed locally around cracks rather than only through global scalar error.

partial observation? Second, can future crack growth be predicted from the identified state? Third, does uncertainty coverage remain valid in the crack region when the initial state is

uncertain? This separation is necessary because a poor prediction may come from state non-identifiability, surrogate error, registration error or physics-model mismatch.

WP3.2: Identification-to-prediction comparison. The second task is to compare prediction strategies under the same partial-data conditions. A direct baseline will predict future damage from crack masks or geometry channels alone. An identification-informed route will first infer a hidden state and uncertain inputs using WP1, then pass them to the WP2 surrogate. A hybrid route will add weak physics constraints or physics-informed features. The key question is whether explicit identification improves prediction compared with direct image-to-field or mask-to-field prediction. The validation metrics will include hidden-field reconstruction error, parameter-recovery error, crack-tip prediction error, local damage-field error, fatigue-history error where available, active-driver error where available, crack-region uncertainty coverage, rollout stability and physical-violation counts. These metrics preserve the Chapter 4 lesson: scalar agreement is not enough, and local hidden fields must be audited.

WP3.3: Hybrid physics/data ablations. The third task is to test whether physics helps once data are available. The comparison will include at least three levels. The first level is data-supervised prediction without explicit physics terms. The second level adds weak constraints that require only damage-like outputs, such as bounded damage, damage irreversibility, crack-surface energy regularity and process-zone smoothness. The third level, if richer fields are available or regenerated from FEM, adds stronger terms such as elastic-equilibrium residuals, phase-field residuals, fatigue-history update consistency and active-driver consistency. This staged design avoids the main weakness of the first-year PIDL route. The model is not asked to find the entire fracture solution from physics loss alone. Instead, data provide the main supervision, and physics is tested as an incremental source of consistency, regularisation and uncertainty control. If physics improves identification or rollout reliability, it becomes part of the final framework. If it does not, the negative result is still informative because it identifies which physical constraints are not useful under the available data and model class.

WP3.4: Uncertainty propagation and decision-level outputs. The final task is to propagate uncertainty through the full chain. The prediction will not be reported as a single deterministic crack path. It will include uncertainty from registration, state identification, material/loading parameters and surrogate model error. Maintenance-relevant indicators can then be derived cautiously, such as probability of crack extension beyond a threshold, expected crack-tip movement, widening risk or uncertainty around an active damage zone.

These indicators will not be presented as a fully validated remaining-useful-life map unless field-scale validation becomes available. The expected output is a locally calibrated crack-growth prediction prototype under partial observations.

Table 5.3 Hybrid validation structure in Work Package 3.

Validation route	Main test	Expected evidence
Direct prediction baseline	Predict future damage or crack field directly from crack mask and geometry channels	Shows the performance limit without explicit hidden-state identification
Identification-informed prediction	Infer hidden state and uncertain inputs before rollout through the supervised surrogate	Tests whether identification improves future prediction and uncertainty calibration
Weak hybrid model	Add damage bounds, irreversibility and crack-surface regularity to data supervision	Tests whether low-field physics constraints improve reliability when only damage labels are available
Strong hybrid model	Add equilibrium, phase-field, history-update or fatigue-consistency terms where fields are available	Tests whether richer physics improves hidden-field recovery and rollout consistency
Repeated-observation validation	Compare predicted crack changes against later inspection observations	Provides a practical validation route when full hidden FEM fields are unavailable

The deliverable of WP3 is a hybrid validation framework for partial-data crack-growth prediction. It closes the loop between inspection outputs, hidden-state identification, supervised prediction and field-level reliability checks. It also defines the final role of PIDL-derived physics: not as a pure standalone predictor, but as a set of constraints and diagnostics that may improve a data-supervised pipeline.

5.3 Research Programme and Risk Assessment

The revised programme is organised around a minimum viable thesis rather than a field-ready deployment promise. The main completion risk is that the full route from inspection data to hidden-state identification, calibrated neural prediction and field-scale pavement validation is too large to complete at full depth within one PhD. The work packages are therefore staged so that each one can produce a defensible contribution even if later field-scale validation is limited. The minimum deliverables are: a data-supported identification pipeline tested on synthetic or semi-synthetic observations; a supervised phase-field neural-operator study with crack-region uncertainty calibration; and a hybrid partial-data validation study that tests the identification–prediction chain. These deliverables answer the central research question by showing why physics-only supervision is insufficient, how data can support hidden fracture-state identification, and how identified states can be propagated for future

prediction under uncertainty. The first technical risk is data availability. Real pavement observations may not include repeated views, reliable registration, crack width, structural proxies or loading histories. This risk will be managed by starting with synthetic degradation of FEM fields, then moving to semi-synthetic or open inspection data where possible. The claim will be limited according to the available data.

The second technical risk is non-identifiability. Visible crack geometry alone may not uniquely determine internal damage, fatigue history, material degradation or loading history. The programme treats this as a research output rather than as a failure. WP1 and WP3 will quantify identifiability limits and propagate uncertainty instead of reconstructing a single deterministic hidden state. The third technical risk is surrogate reliability. A neural operator may achieve low global error while remaining overconfident near cracks. WP2 therefore uses crack-region calibration, coverage, rollout stability and local field metrics as primary reliability gates. These metrics inherit the field-level validation logic established in Chapter 4.

The fourth technical risk is physics-regularisation failure. Physics-informed losses may not improve prediction if the available fields are insufficient or if the physics terms are poorly balanced. For this reason, physics-informed extensions are placed after supervised baselines and are treated as ablations. If they improve identification or rollout reliability, they become part of the final framework. If they do not, their contribution remains diagnostic: they define which physical assumptions fail to add value under the available data.

5.4 Chapter Summary

This chapter has revised the future research plan around the main lesson from Chapters 3 and 4. The current PIDL formulation, trained without data supervision, is useful as a diagnostic tool but is not yet reliable as a field-level crack-growth predictor. This supports a data-centred future direction. The first future task is identification: estimating crack-growth initial states and uncertain material or loading inputs from partial observations. The second task is prediction: learning calibrated supervised neural operators that propagate identified states into future damage fields. The third task is hybrid validation: testing the full identification–prediction chain under partial data and assessing whether physics-informed constraints improve reliability once data supervision is available. In this revised structure, the original pure-PIDL work package is not continued as the main future route. Its useful components are retained as physics requirements, field-level validation gates and optional regularisation terms. This makes the thesis more coherent: Chapter 4 explains why data are needed, while Chapter 5 proposes how data can be used for identification and prediction.

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Appendix A

Pre-final-alignment PIDL Method Trials

A.1 Purpose and Scope

This appendix records the main physics-informed deep learning (PIDL) method trials carried out before the final alignment protocol was fixed. The purpose is not to repeat the Chapter 4 result narrative, but to show how the exploratory phase moved from scalar feasibility checks to stricter field-level validation. These trials are therefore reported as a method route rather than as a catalogue of archive paths. Full run identifiers, scripts, commits, logs and machine-specific paths are retained in the internal project inventory, but are not reproduced here because they would make the appendix less readable for an external reader.

A.2 Method Route

The exploratory work followed the route shown in Fig. A.1. The first question was whether PIDL could reproduce plausible scalar fatigue lives. Once that was possible, the work tested increasingly specific explanations for the remaining field mismatch: insufficient representation, insufficient local attention, incorrect fatigue-history or driver feedback, and finally avoidable differences between the FEM and PIDL reference states. This sequence matters because the negative trials are not isolated failures. They explain why the later comparison had to be based on strict alignment, state mapping and active-driver/process-zone metrics rather than on scalar N_f alone.

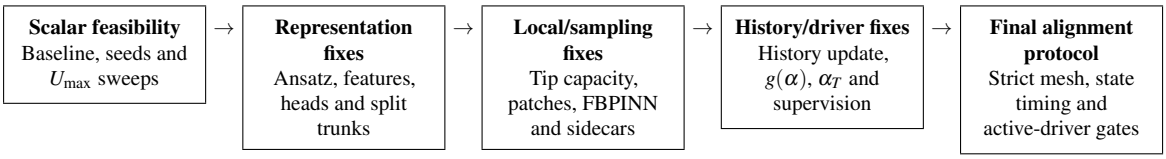


Fig. A.1 Method route used to organise the PIDL trials before the final alignment protocol. The route records the order in which hypotheses were tested, not a claim that each step solved the field mismatch.

A.3 Method Families Tested

Table A.1 summarises the main method families along this route. The table keeps the discussion at the level of ideas, because the detailed run names are useful for audit but not for reading. The main lesson is that many interventions shifted event timing, crack-path appearance or local scalar values, but they did not by themselves establish FEM-like active-driver and fatigue-history evolution. That outcome motivated the shift from adding more PIDL variants to first fixing the comparison protocol.

Table A.1 Pre-final-alignment PIDL method families, organised by the exploratory route rather than by raw run identifier.

Route stage	Why it was tried	Representative changes	What was learned
Scalar feasibility	Establish whether the PIDL formulation could reproduce plausible fatigue-life trends before judging local fields.	Baseline runs, multi-seed checks, $U_{\max}$ sweeps and early scalar comparisons against FEM.	Scalar N_f agreement was achievable in several settings, but it was not sufficient evidence of a correct hidden field.
Representation fixes: basis enrichment	Test whether the crack-tip field was being smoothed because the coordinate network lacked suitable local basis functions.	Williams-style features, Fourier features, Mode-I enrichment and enriched ansatz variants.	These variants changed some scalar or local quantities, but they did not reliably recover the FEM-scale process-zone driver.
Representation fixes: boundary and symmetry constraints	Remove errors that should be exact properties of the benchmark rather than quantities discovered by optimisation.	Hard symmetry constraints and exact side-boundary distance-function ansatz variants.	Encoding known boundary structure helped remove avoidable errors, but it did not by itself prove field-mechanism alignment.
Representation fixes: output factorisation	Test whether shared network representations or optimiser ordering caused the displacement and damage fields to interfere with one another.	Multi-head models, staged u/v-then-α schedules, u/v-only and α-only branches, no-joint variants and true split-trunk networks.	Split-trunk and staged-head runs changed event timing or history magnitude, but the active-driver/history trajectory remained non-FEM-like.
Representation fixes: discontinuity-aware kinematics	Test whether the crack representation needed an explicit discontinuity or distance-to-crack input rather than a smooth coordinate-only field.	XFEM jump heads, SDF ribbons and discontinuity-aware u/v-correction branches.	Boundary-event timing changed in some runs, but the change did not establish a corrected process-zone mechanism.
Local/sampling fixes: local capacity	Test whether the global network needed a local model near the crack tip to represent steep gradients.	Tip-local MLP, Fourier and SIREN heads, additive local patches and FBPINN-style overlapping local subnetworks.	Local capacity shifted event timing and propagation mode, but the fields often remained centreline- or boundary-dominated rather than FEM-like.
Local/sampling fixes: adaptive sampling and sidecars	Test whether the process zone was under-sampled in the collocation set while keeping the physics objective conceptually fixed.	Static-tip oversampling, adaptive sidecars, cumulative tip-following, score-driven refinement and mesh-transport diagnostics.	Extra crack-tip samples lifted some early history metrics and shifted propagation in some seeds, but the effect was modest and did not close the active-driver/history gap.
Local/sampling fixes: loss and history weighting	Test whether the history term or process-zone terms were under-weighted relative to the energy terms during training.	Tip weighting, adaptive λ_{hist}, FI-PINN-style reweighting and related history-loss balancing diagnostics.	Reweighting could shift timing or inflate history, but it either did not become an active balancing mechanism or risked changing the variational objective.

continued on next page

Table A.1 Pre-final-alignment PIDL method families (continued).

Route stage	Why it was tried	Representative changes	What was learned
History/driver fixes: fatigue update variants	Test whether the mismatch came from the fatigue-history update, degradation law or scalar fatigue parameter rather than from representation alone.	Spatial α_T, log-f variants, raw-driver and lagged-driver variants, hard irreversibility, residual stiffness and inverse α_T.	These runs could shift N_f, inflate history or advance fracture, but scalar compensation did not repair the active $g(\alpha)\psi^+$ field.
History/driver fixes: supervised and oracle probes	Separate the learned-network limitation from the downstream fatigue accumulator by injecting or supervising selected hidden quantities.	Weak ψ^+ supervision, local α supervision, direct field supervision, ψ^+ hacks and oracle driver injection.	FEM-like upstream driver information was sufficient to force strong accumulation, but direct supervision was not a clean pure-physics closure tool without normalisation and gating.
Final alignment protocol	Remove avoidable comparison ambiguity before interpreting the remaining mechanism gap.	Strict FEM-mesh and soft-hist0 references, state-zero/state-timing exports, common-probe comparisons and raw/active-driver separation.	The project moved from “try another variant” to a stricter rule: new PIDL sweeps must pass active-driver and process-zone gates, not only scalar N_f.

A.4 Detailed Method Notes

The following notes expand the compact inventory in Table A.1. Each method card follows the same logic: why the method was tried, what changed in the model or training protocol, what was observed, why the result was not retained as a closure claim, and how the observation changed the next step. The figures are included to make the trial outcomes readable. They are trial evidence and transition evidence, not a second presentation of the Chapter 4 final aligned result.

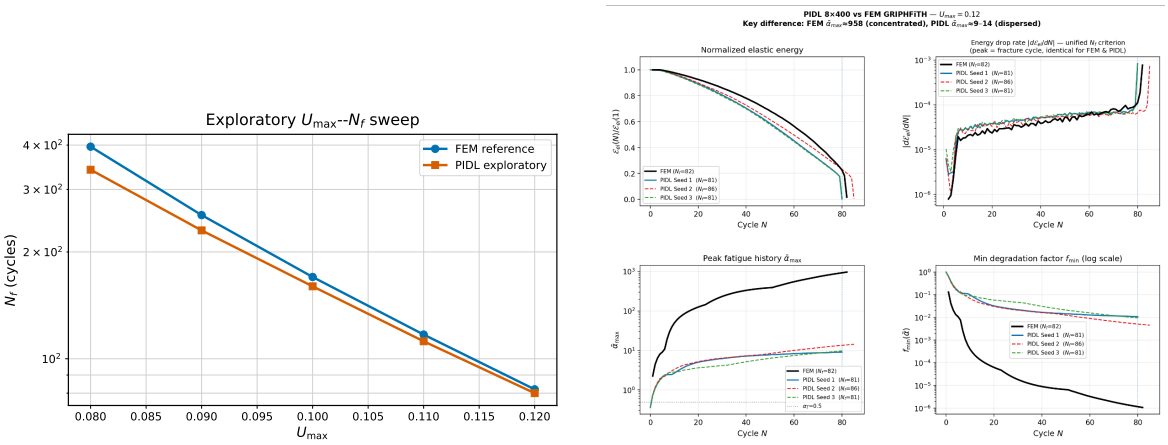
A.4.1 Scalar Feasibility Before Field Validation

Reason. The first question was whether PIDL contained any useful fatigue-life signal before detailed field diagnostics were attempted. A model that failed at the fatigue-life scale would not justify more expensive process-zone comparisons. The scalar check used the relative fatigue-life error

$$e_N = \frac{N_f^{\mathrm{PIDL}} - N_f^{\mathrm{FEM}}}{N_f^{\mathrm{FEM}}},$$

as the entry condition for the later trials.

Method. The early runs varied $U_{\max}$ within the same single-edge notched tension benchmark and compared the trend of N_f with load amplitude. At this stage the comparison was deliberately scalar. Hidden fields such as damage, fatigue history and degradation were inspected only after the fatigue-life trend looked plausible enough to continue.



(a) Scalar $U_{\max}$–N_f trend.

(b) Hidden-field discrepancy at $U_{\max} = 0.12$.

Fig. A.2 Early scalar feasibility and the field-level warning that followed it. The scalar trend was plausible, but the internal fatigue-history and degradation fields were not FEM-like.

Result. Figure A.2 shows why the scalar check was useful but insufficient. The $U_{\max}$–N_f trend was monotone and close enough to justify further work. However, at $U_{\max} = 0.12$, the FEM history peak was about two orders of magnitude larger than the PIDL history peak in the pre-alignment diagnostic, and the spatial degradation pattern was different.

Failure reading. The scalar result did not fail as a feasibility test. It failed as a validation criterion. A model can reach a similar fracture-cycle indicator while distributing fatigue history over the wrong region.

Next adjustment. The next step was to test whether this hidden-field gap came from the neural representation. That led to Williams-style features, Fourier features, enriched outputs and later local correction heads.

A.4.2 Representation Fixes

Williams, Fourier and Mode-I Enrichment

Reason. The first representation route tested whether the coordinate MLP was smoothing the crack-tip fields. The concern was spectral bias: a smooth neural field might reproduce broad displacement trends while failing to represent the sharp tensile-energy and fatigue-history concentration around the crack tip.

Method. Three enrichment forms were tested. Williams-style inputs supplied crack-tip radius and angular functions,

$$\eta_W = \left(\sqrt{r/\ell_0},\, r/\ell_0,\, \sin\frac{\theta}{2},\, \cos\frac{\theta}{2},\, \sin\frac{3\theta}{2},\, \cos\frac{3\theta}{2} \right).$$

Fourier features replaced the raw coordinate input by

$$\gamma_B(\mathbf{x}) = [\sin(2\pi B\mathbf{x}),\, \cos(2\pi B\mathbf{x})]\,,$$

where B was fixed after initialisation. The Mode-I enriched-output route added an analytic crack-tip term to a smooth neural residual, so that the network did not have to learn the whole near-tip form from coordinates alone.

Result. Figure A.3 shows the main pattern. Williams features reached an event cycle near the scalar window but reduced the history peak relative to the baseline. Fourier features and

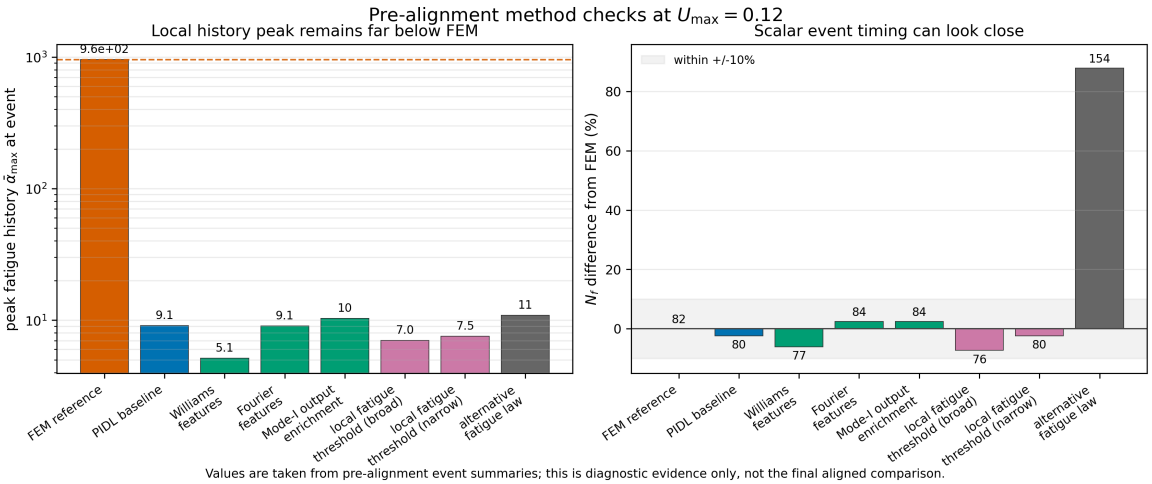


Fig. A.3 Pre-final-alignment representation and fatigue-chain checks at $U_{\max} = 0.12$. Williams, Fourier and Mode-I output enrichment could move event timing, but the local fatigue-history peak remained far below the FEM reference in these pre-alignment trials.

Mode-I enrichment kept event timing close to the FEM reference in some runs, but their history peaks still stayed far below the FEM scale.

Failure reading. Representation enrichment was not useless; it showed that local basis choice affected the result. The failure was narrower: richer inputs or enriched outputs did not force the active driving field and fatigue-history accumulator to co-locate in the FEM process zone. The moving Williams anchor also made the peak less stationary, which is harmful for a cumulative history variable.

Next adjustment. The next route separated two issues that enrichment mixed together: whether exact known structure should be imposed by construction, and whether the displacement and damage outputs needed independent authority.

Boundary and Symmetry Constraints

Reason. Some errors in the early runs were not quantities that the optimiser should have to discover. The benchmark is mirror-symmetric about the crack line, and the side boundaries are traction-free. This motivated hard symmetry and exact-boundary ansatz variants as guardrail tests.

Method. The hard-symmetry route encoded the mirror structure directly in the field transformation, rather than penalising the symmetry error after prediction. The exact-boundary

route used distance-function factors so that selected side-boundary residuals vanish by construction. These changes were treated as representation constraints, not as new fatigue physics.

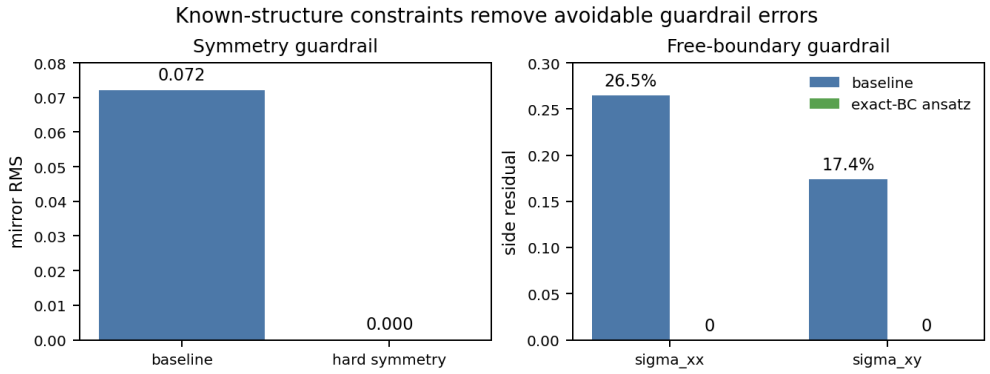


Fig. A.4 Guardrail effect of known-structure constraints. Hard symmetry removes the mirror-RMS error by construction, while the exact-boundary ansatz removes the measured side-boundary residuals in the diagnostic setting. These checks remove avoidable errors but do not by themselves establish FEM-like fatigue-history evolution.

Result. Figure A.4 records the guardrail result. The baseline mirror-RMS error was about 0.072, while the hard-symmetry construction reduced it to numerical zero. The exact-boundary ansatz similarly removed the side residuals measured in the diagnostic setting.

Failure reading. This route succeeded for the guardrail quantities but did not close the fracture mechanism. Enforcing a known symmetry or boundary condition can remove an avoidable error while the active driver and history field remain non-FEM-like.

Next adjustment. The result justified separating two classes of changes: known mechanical structure should be imposed where appropriate, while crack-tip localisation still required direct tests of output coupling and local capacity.

Output Factorisation

Reason. The next hypothesis was that the global shared network made the displacement and damage fields interfere with one another during optimisation. This motivated multi-head models, staged output-head schedules and a true split-trunk run.

Method. The staged-head tests trained u/v rows and the α row separately before a joint solve. The true split-trunk form used independent subnetworks,

$$(u_\theta, v_\theta) = \mathcal{N}_{uv}(\mathbf{x}; \theta_{uv}), \qquad \alpha_\theta = \mathcal{N}_\alpha(\mathbf{x}; \theta_\alpha),$$

instead of one shared trunk for all outputs. These were optimisation-path and representation-partition tests; the variational energy and fatigue law were not changed.

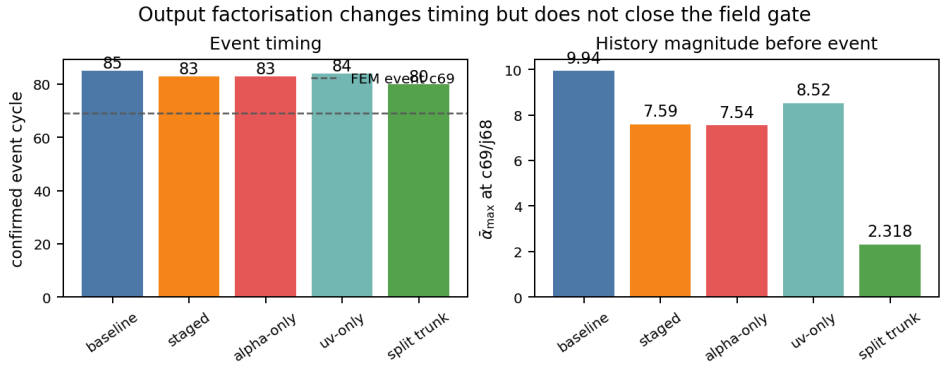


Fig. A.5 Output-factorisation discriminator under the FEM-mesh soft-hist0 setting. Staged heads and a true split trunk changed event timing and the cycle-69 history magnitude, but they did not produce a FEM-like active-driver/history trajectory.

Result. Figure A.5 summarises the scalar discriminator. The staged branches confirmed around cycles 83–84, while the true split-trunk branch confirmed around cycle 80. At the fixed $c69/j68$ comparison, the staged variants remained near the baseline history scale, and the split-trunk branch reduced $\bar{\alpha}_{\max}$ to about 2.3.

Failure reading. The result closed the simple shared-trunk explanation. Splitting the outputs or changing the per-cycle training order changed event timing, but it did not recover the missing active process-zone driver. The no-joint staged branch was also pathological: it produced large history without a physically useful propagation path.

Next adjustment. The next route gave the crack-tip region local authority rather than only changing the global output schedule.

Discontinuity-aware Inputs

Reason. The discontinuity-aware route tested whether a smooth coordinate-only representation was the wrong kinematic prior for a cracked specimen. The idea was that distance-

to-crack or jump-style features might help the displacement field represent the crack more cleanly.

Method. Two strict-ish endpoint variants were retained. The SDF-ribbon branch supplied a signed-distance ribbon feature near the crack. The XFEM-style branch added a jump-aware u/v correction, schematically

$$u_\theta(\mathbf{x}) = u_{\mathrm{NN}}(\mathbf{x}) + \mathscr{H}(\mathbf{x})v_{\mathrm{NN}}(\mathbf{x}),$$

restricted to a displacement correction rather than a full new damage model.

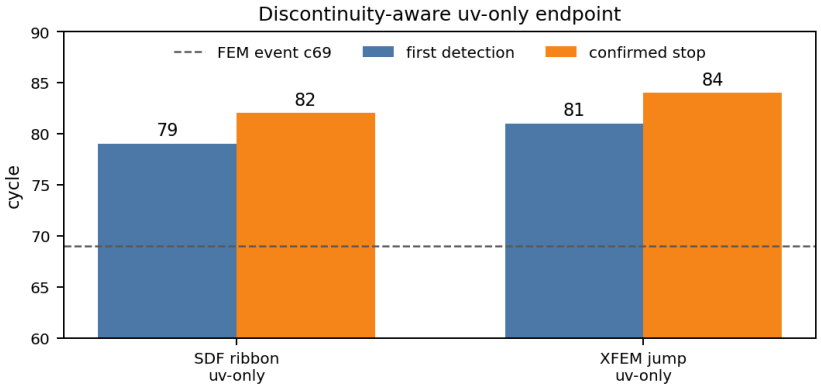


Fig. A.6 Strict-ish SDF and XFEM uv-only endpoints. The discontinuity-aware inputs shifted boundary-event timing, but the retained endpoint does not include a completed active-driver/process-zone field comparison showing mechanism closure.

Result. Figure A.6 shows the retained endpoint: the SDF-ribbon branch first detected a boundary event at about cycle 79 and confirmed at cycle 82, while the XFEM-style branch first detected at about cycle 81 and confirmed at cycle 84.

Failure reading. The result is negative but diagnostic. The event window changed, but the available evidence did not show that either discontinuity feature repaired the active-driver and process-zone mechanism. The older SDF archive predates strict setting alignment and should therefore be treated only as lineage, not as a separate final evidence point.

Next adjustment. No new SDF/XFEM sweep should be launched unless it names the active-driver/process-zone metric it is intended to improve.

A.4.3 Local and Sampling Fixes

Tip-local Heads

Reason. The tip-local route tested whether the global network needed a compact correction model near the crack tip to represent steep gradients. This was a local-capacity hypothesis rather than a change to the fatigue law.

Method. The tested local heads used MLP, Fourier and SIREN forms in static or moving windows. The local correction can be written schematically as

$$q_\theta(\mathbf{x}) = q_g(\mathbf{x}) + \chi_r(\mathbf{x})\, q_l(\mathbf{x}),$$

where χ_r gates the local head around the crack-tip region.

Result. The tip-local matrix in Fig. A.7 shows that many variants fractured in a plausible scalar window. However, their final α fields still looked centreline- or boundary-dominated rather than like the FEM process-zone envelope.

Failure reading. The local head changed the path to fracture but did not repair the hidden mechanism. A compact local correction can affect morphology without making the local fatigue-history and active-driver fields FEM-like.

Next adjustment. The next adjustment strengthened local authority using additive local patches and FBPINN-style overlapping subdomains.

Local Patches and FBPINN-style Decomposition

Reason. The local-patch and FBPINN branches tested whether the tip-local head failed because it was too small or too weakly coupled to the main field. The aim was to give local subnetworks real authority while keeping the same variational energy and fatigue update.

Method. The additive patch kept a global field and added a windowed local correction. The FBPINN-style form blended overlapping local subnetworks,

$$q_\theta(\mathbf{x}) = (1-\beta(\mathbf{x}))q_g(\mathbf{x}) + \beta(\mathbf{x})\frac{\sum_i \chi_i(\mathbf{x})q_i(\mathbf{x})}{\sum_i \chi_i(\mathbf{x})},$$

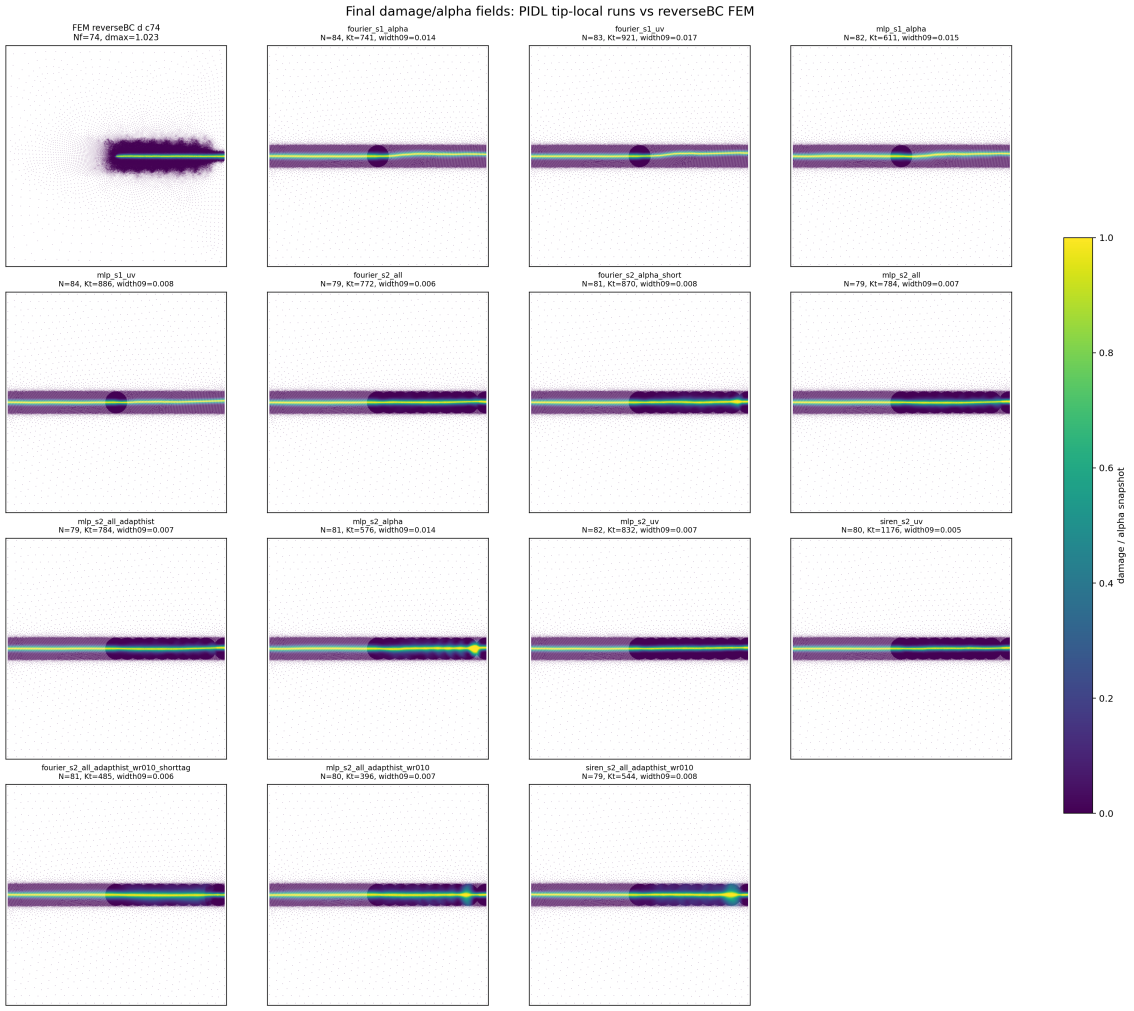


Fig. A.7 Tip-local representation matrix against the reverseBC FEM reference. The tested MLP, Fourier and SIREN local heads changed event timing and local appearance, but their final fields remained more centreline- or boundary-dominated than the FEM process-zone envelope.

so that local models carried the raw field inside their windows rather than merely adding a small residual.

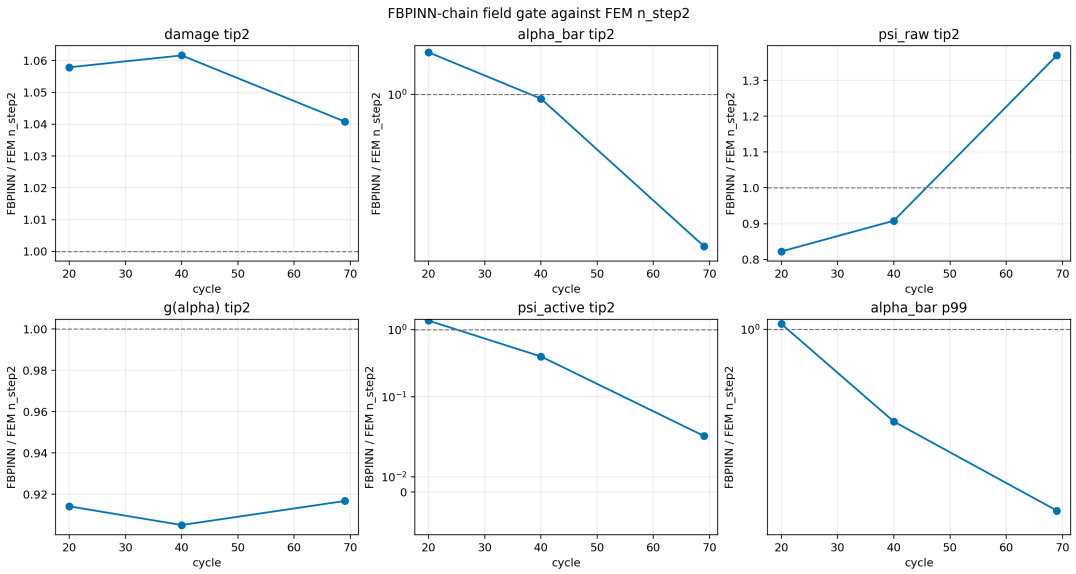


Fig. A.8 FBPINN-chain field gate against the FEM $n_{\text{step}} = 2$ reference. The local subdomain model kept damage amplitude close to FEM in the sampled tip region and avoided the usual right-boundary stop by the final saved state, but its active driver and high-percentile history still fell below the FEM process-zone scale.

Result. The additive local patch was baseline-like or worse: all-output and α-only patches confirmed around cycles 83–84 with final $\bar{\alpha}_{\max}$ near the baseline scale, while the u/v-only patch produced much lower history. The FBPINN chain changed the failure mode more clearly. It did not hit the right boundary by the final saved state, and Fig. A.8 shows near-tip damage ratios close to the FEM reference, but the $c69$ active-driver tip ratio was only about 0.037.

Failure reading. The result showed that local representation authority matters, because FBPINN changed the propagation mode. It also showed that local authority alone is not closure: the active driver remained too small or off-location in the process zone.

Next adjustment. Future local/domain-decomposed designs should be promoted only if they improve the active-driver and process-zone field gate, not merely event timing or damage amplitude.

Adaptive Sampling and Sidecar Refinement

Reason. The adaptive-sampling route asked whether the process zone was under-represented in the collocation set. This was a narrower hypothesis than loss reweighting: the aim was to change where training information came from while keeping the Deep Ritz and Carrara objective conceptually fixed.

Method. The sidecar branch changed the sampling distribution,

$$\mathbf{x}_i \sim p_\eta(\mathbf{x}),$$

where p_η concentrates more points near the current or expected crack-tip corridor. The static-tip S1 sidecar oversampled a fixed crack-tip region. The S2 variants added cumulative tip-following or score-driven refinement, together with mesh-transport checks so that history variables were not accidentally compared across incompatible point sets.

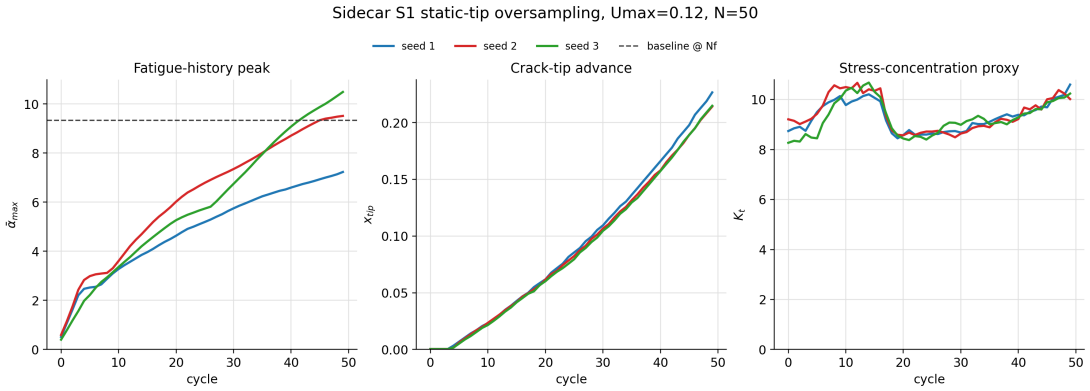


Fig. A.9 Static-tip adaptive-sampling sidecar across three seeds. The sidecar increased early fatigue-history accumulation in some seeds while preserving the stress-concentration proxy, but the three-seed spread and the limited horizon show that sampling alone was not a closed mechanism result.

Result. Figure A.9 shows the useful part of the test. By cycle 49, the S1 sidecar reached $\bar{\alpha}_{\max} = 7.23$, 9.52 and 10.49 across three seeds, with crack-tip positions near $x = 0.21$–0.23 and K_t remaining near 10. The branch therefore supported the idea that sampling dilution contributed to the early history gap.

Failure reading. The route did not close the FEM mechanism. The gain was seed-dependent, the horizon was shorter than the final fracture comparisons, and the branch

did not demonstrate a corrected active-driver and process-zone history field. It was a useful diagnostic, not a validation endpoint.

Next adjustment. Adaptive sampling should remain a placement tool. Future sampling variants should be judged by common-probe active-driver and process-zone metrics, not by early $\bar{\alpha}_{\max}$ lift alone.

Loss and History Weighting

Reason. The weighting route tested a different hypothesis: the process zone might have been present in the samples, but the fatigue-history term or local process-zone residuals might have been too weak in the objective.

Method. These variants changed the objective through weights such as

$$\mathscr{L} = \mathscr{L}_{\text{phys}} + \lambda_{\text{hist}} \mathscr{L}_{\text{hist}},$$

or through FI-PINN-style reweighting. This route was kept separate from adaptive sampling because changing $p_\eta(\mathbf{x})$ changes the training measure, whereas changing λ_{hist} changes the loss balance and can alter the effective variational problem.

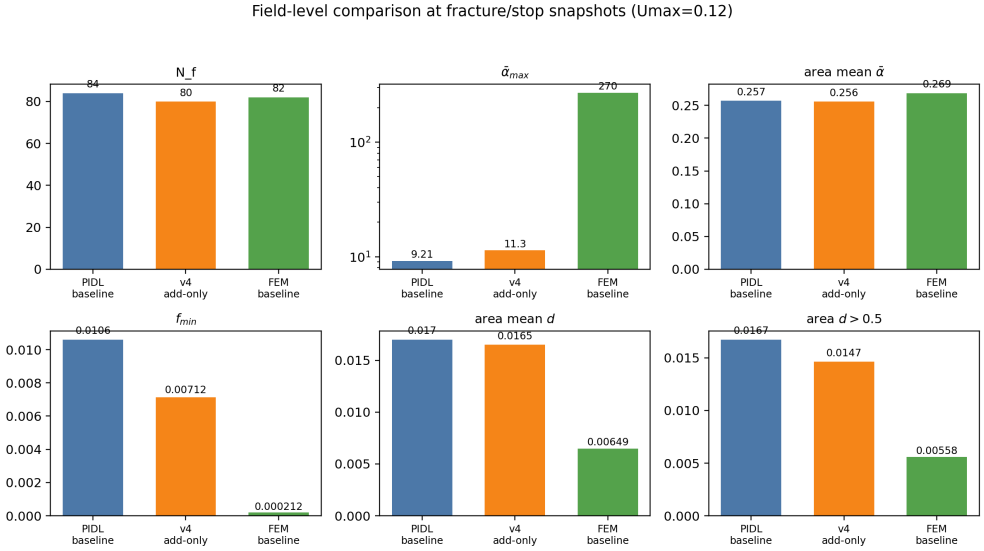


Fig. A.10 Adaptive λ_{hist} and v4 add-only refinement aggregate comparison at fracture or stop snapshots. The variants shifted event timing and some aggregate quantities, but the FEM reference still had a much larger peak fatigue-history scale than either PIDL variant in this pre-alignment comparison.

Result. The adaptive-history and v4 add-only refinement runs fractured, but Fig. A.10 shows that their aggregate agreement remained limited. The v4 add-only variant confirmed around cycle 90 and the baseline adaptive-history run around cycle 92, while the peak $\bar{\alpha}$ remained far below the FEM value. The run logs also showed $\lambda_{\text{hist}} = 1.0$ late in the trajectory, so the adaptive loss did not become an active balancing mechanism.

Failure reading. The failure was not that history weighting is irrelevant. It was that the tested weighting mechanisms did not correct the active-driver/history mechanism, and more aggressive weighting would risk changing the problem being solved rather than only improving resolution.

Next adjustment. The route suggested a cleaner future distinction: sampling changes should be used to place points, while loss changes should be treated as a separate modelling choice with explicit pass rules for active-driver, fatigue-history and process-zone fields.

A.4.4 History, Driver and Supervised Probes

Fatigue-history and Driver Variants

Reason. After representation and sampling changes failed to close the field gap, the next hypothesis was that the fatigue-history update or degraded active driver was the main bottleneck. This route asked whether the problem sat upstream in raw tensile energy, downstream in the Carrara-style accumulator, or in the coupling between damage, degradation and history.

Method. The key diagnostic distinction was between raw tensile energy and the active driver,

$$\psi_{\text{active}} = g(\alpha)\psi^{+}.$$

The discriminator compared the current active driver, a lagged-degradation driver and a raw-driver ceiling. Other fatigue-chain variants tested spatial or inverse α_T, residual stiffness, hard irreversibility and alternative degradation forms.

Result. Figure A.11 shows the key negative. The raw-driver branch made the history magnitude extremely large, and the lagged-degradation branch also increased accumulated history relative to the current-active branch. However, by the FEM event cycle the active-driver ratio in the tip region stayed near 0.02 for all three modes.

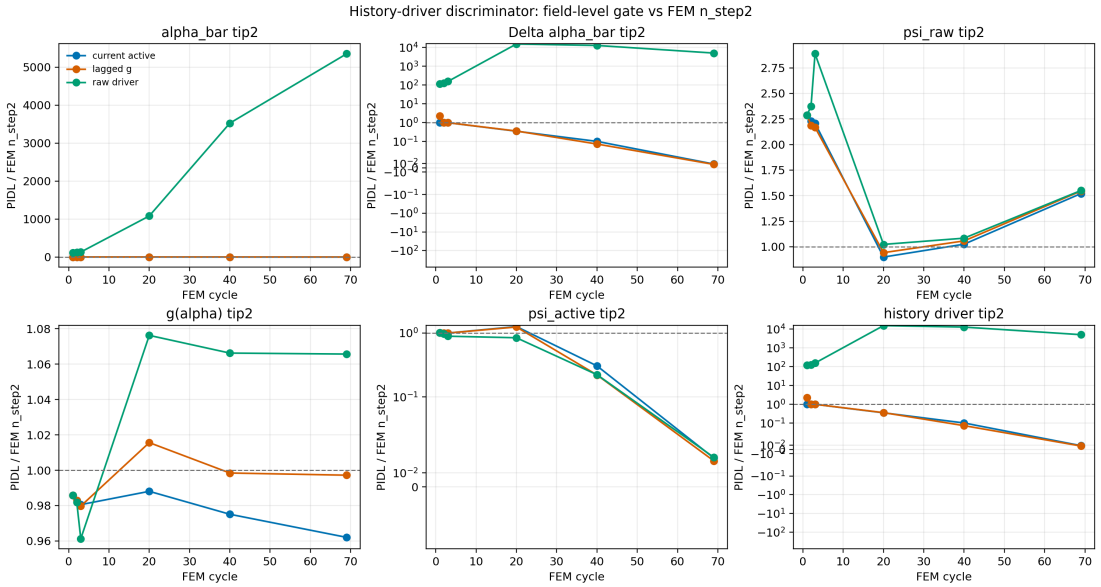


Fig. A.11 History-driver discriminator against the FEM $n_{\text{step}} = 2$ reference. Raw-driver and lagged-driver variants can inflate accumulated history, but the active $g(\alpha)\psi^{+}$ process-zone ratio remains far below FEM late in the trajectory.

Failure reading. This is scalar compensation rather than mechanism closure. Making the history clock larger can change event timing or $\bar{\alpha}$ magnitude, but it does not by itself reconstruct the degraded active process-zone driver.

Next adjustment. Future driver changes must be evaluated by raw and active driver fields together. A large $\bar{\alpha}$ value is not enough unless $g(\alpha)\psi^{+}$, $\Delta\bar{\alpha}$ and damage growth move toward the FEM process zone together.

Weak, Direct-field and Oracle Probes

Reason. The supervision and oracle route separated a learned-field limitation from a downstream accumulator limitation. If FEM-like upstream information could force stronger history accumulation, then the accumulator was at least capable of responding; if direct supervision failed, the problem was not solved by simply adding a target.

Method. Weak and direct supervision added target losses of the form

$$\mathcal{L}_{\text{sup}} = \lambda_s \| s_\theta - s_{\text{FEM}} \|_2^2,$$

with s chosen from fields such as α, $\bar{\alpha}$, ψ_{raw} or ψ_{active}. Oracle probes instead replaced or injected selected FEM or constructed hidden quantities into the driver/history path. The oracle route was diagnostic, not a deployable PIDL method.

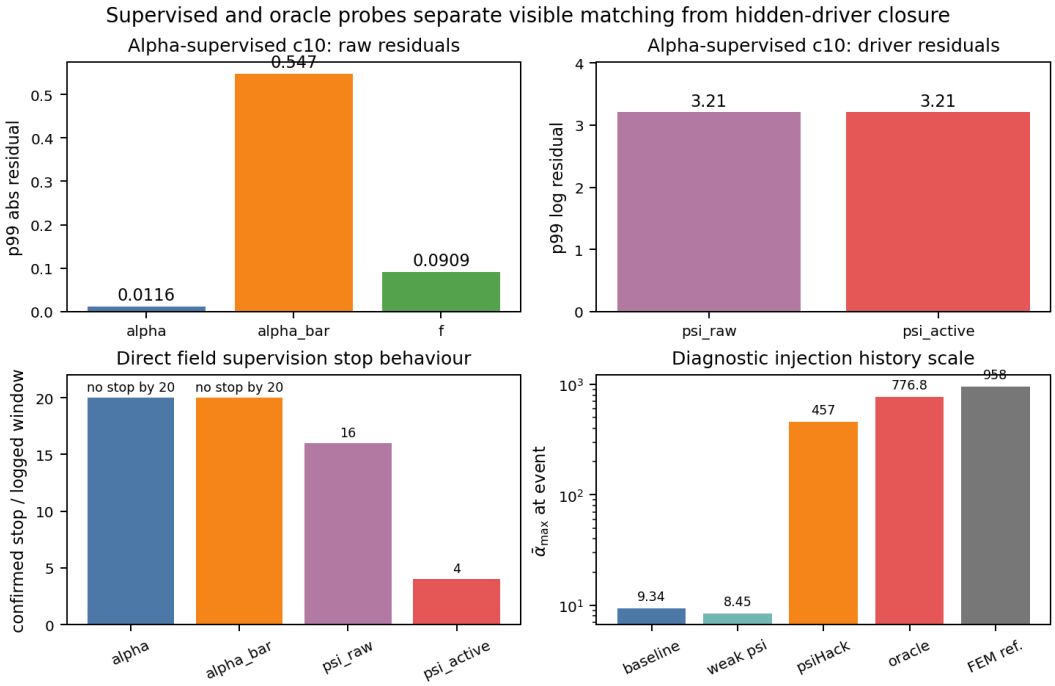


Fig. A.12 Supervised and oracle probe summary. Alpha supervision could keep visible damage residuals small in early sampled states while hidden history and driver residuals remained large. Ungated hidden-driver supervision caused premature boundary failure, whereas diagnostic injection showed that sharp upstream information can produce FEM-scale history.

Result. Figure A.12 summarises the split. Weak ψ^{+} supervision increased the target driver but did not lift $\bar{\alpha}_{\max}$ beyond the baseline scale. Direct ψ_{active} supervision was too aggressive and confirmed boundary failure by about cycle 4; direct ψ_{raw} supervision confirmed by about cycle 16. Alpha supervision kept the visible damage residual small in the early sampled states, but $\bar{\alpha}$ and the driver fields remained far from FEM. Oracle and psiHack-style injections showed the opposite diagnostic: sufficiently sharp upstream information can produce much larger history accumulation.

Failure reading. This route did not fail as a diagnostic. It showed that the accumulator can respond if the upstream driver is sharp enough. It failed as a final method because

oracle injection uses FEM information by construction, and ungated direct supervision either matched the wrong visible quantity or destabilised the boundary behaviour.

Next adjustment. Direct field supervision should not be repeated as a long production run until the proposal specifies target normalisation, a gating schedule, early residual tables, and process-zone/active-driver pass rules.

Appendix B

Research Programme

Table B.1 gives an indicative programme for the remaining research. The schedule is intended to show how the work will be kept within a feasible thesis scope rather than to fix every experiment to a rigid calendar. The main structure is a minimum viable thesis route: close the FEM–PIDL diagnostic benchmark, complete a supervised phase-field surrogate and calibration audit, and demonstrate an inspection-informed prototype on synthetic or degraded observations. Physics-aware surrogate extensions and field-scale pavement validation are treated as decision-gated depth rather than as mandatory completion requirements.

Table B.1 Indicative research programme for the remaining work. Coloured bars mark the main period of activity for each work package, while milestone labels indicate completion gates, scope decisions and planned thesis outputs.

Work package and activity		2026		2027				2028	
		Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
WP1 PIDL diagnosis	Close FEM–PIDL validation protocol and field-level gates								
	Test model-boundary variants and synthetic inverse cases								
	Decide PIDL role as predictor, diagnostic or constraint source								
WP2 Supervised surrogate	Neural-operator and CA-FNO calibration audit								
	Fatigue trajectory dataset and multi-step rollout tests								
	Physics-aware extension, if fields and scope allow								
WP3 Inspection prototype	Synthetic inspection-to-state mapping								
	Uncertainty-aware surrogate rollout prototype								
Scope Completion gates	Minimum viable thesis and fallback decisions								
Writing Papers and thesis	Publication preparation and thesis integration								
Planned milestones			**M1**		**M2**	**M3**	**M4**	**M5**	

Milestones: M1 = FEM–PIDL validation gates fixed; M2 = supervised surrogate and CA-FNO calibration audit completed; M3 = minimum viable thesis route and PIDL role fixed; M4 = inspection-to-state prototype demonstrated on controlled observations; M5 = thesis scope, publication plan and remaining extensions fixed.